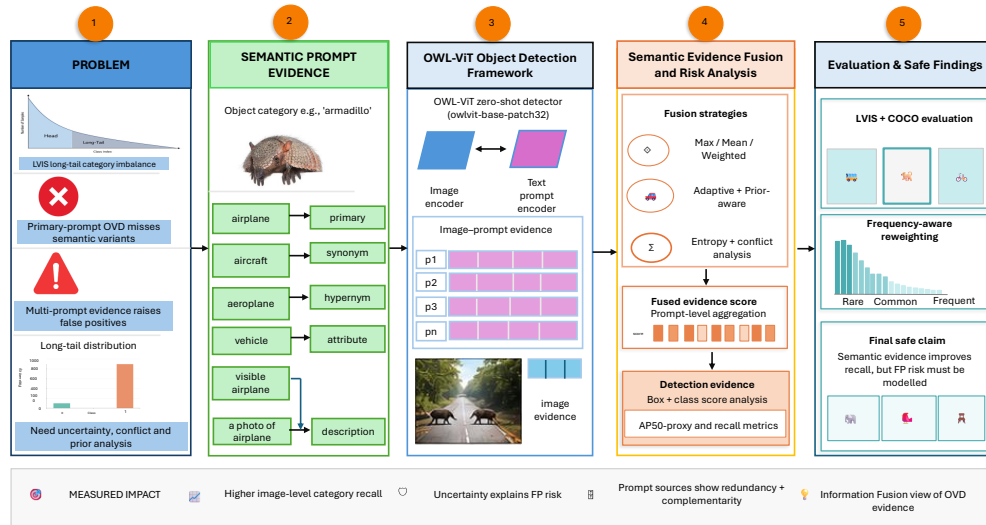


Graphical Abstract

Prior-Aware Multi-Source Semantic Evidence Fusion for Long-Tail Open-Vocabulary Object Detection

Haripriya Chalicheemala, Raja Hashim Ali, Iftikhar Ahmed



Highlights

Prior-Aware Multi-Source Semantic Evidence Fusion for Long-Tail Open-Vocabulary Object Detection

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- A multi-source semantic evidence fusion framework is proposed for long-tail object detection using vision-language models.
- Semantic prompt generation combines primary prompts, WordNet semantic variants, template prompts, and descriptive semantic evidence.
- Several evidence aggregation strategies—including mean fusion, max fusion, weighted fusion, adaptive fusion, and prior-aware fusion—are evaluated.
- Semantic evidence fusion improves recall compared to primary prompt-based detection across the sampled LVIS and COCO evaluation subsets.
- Frequency-aware fusion improves recall for LVIS long-tail categories under the evaluated adaptive threshold protocol.

Prior-Aware Multi-Source Semantic Evidence Fusion for Long-Tail Open-Vocabulary Object Detection

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Abstract

Detecting objects belonging to rarer categories remains challenging because real-world visual datasets exhibit highly imbalanced category distributions. Frequent object categories are represented by numerous visual observations, whereas rare categories often offer only limited semantic and visual cues. Recent vision-language models enable semantic grounding via text prompts; however, most existing approaches rely on a single category prompt and thus fail to adequately leverage complementary semantic information from other linguistic descriptions.

This study investigates semantic evidence fusion as an information fusion problem in which multiple semantically related prompts are treated as correlated sources of semantic evidence. A multi-source framework for semantic evidence is developed using OWL-ViT and category-level semantic prompt generation. Primary prompts, semantic variants derived from WordNet, template prompts, and descriptive semantic formulations are combined using various aggregation mechanisms, including mean fusion, max fusion, weighted fusion, adaptive fusion, and priority-aware fusion.

The proposed framework is evaluated on curated LVIS and COCO subsets using a unified experimental protocol. Six research questions investigate the effectiveness of semantic evidence fusion, the behavior of aggregation strategies, uncertainty and conflict characteristics, redundancy complementarity relationships, the integration of long-tail prior knowledge, and the framework’s overall stability. Experimental results demonstrate that prior-aware semantic evidence fusion consistently improves recall compared to primary prompt baselines. Description based prompts provide the strongest complementary semantic contribution among the prompt sources evaluated, while the analysis of uncertainty and semantic conflicts offers useful diagnos-

tic insights regarding false positives. Frequency aware, prior-guided fusion enhances adaptive thresholding performance for long-tail LVIS categories. Across the evaluated framework variants, Semantic Evidence achieves the highest recall coverage on the selected datasets.

The results demonstrate that the generation of semantic prompts and the aggregation of semantic evidence can be effectively formulated as an information fusion problem, providing practical insights into the integration of semantic evidence for rare object detection.

Keywords: Long-Tail Object Detection, Semantic Evidence Fusion, Information Fusion, Vision Language Models, OWL-ViT, Semantic Prompt Generation, LVIS, COCO

1. Introduction

Object detection is one of the most intensively studied tasks in computer vision and serves as a fundamental component of numerous intelligent systems, including autonomous vehicles, robotics, surveillance, multimedia search, medical image analysis, and human computer interaction. The goal of object detection is to localize and identify semantic objects within visual scenes. Although significant progress has been made through deep learning-based detection architectures, the reliable detection of rare object categories remains an ongoing challenge. Real-world visual datasets typically exhibit highly imbalanced category distributions, where a small number of categories account for the majority of observations, while many categories appear only infrequently. Consequently, object detectors often demonstrate strong performance on frequent categories but lower effectiveness on rare categories.

Recent advances in vision-language models have opened up new possibilities for semantic reasoning within object detection systems. Large-scale vision-language pre-training frameworks such as CLIP, Florence, OWL-ViT, GLIP, and GroundingDINO have demonstrated that text descriptions can provide strong semantic support for open-vocabulary detection and object localization tasks [1, 2, 3, 4, 5]. Architectures like OWL-ViT utilize natural language descriptions as semantic representations and enable category-based visual grounding through text-image matching mechanisms [3]. Unlike conventional detector architectures that require category-specific retraining, vision-language models offer a flexible framework for integrating semantic information directly into the detection process. However, most prompt-based

recognition approaches rely primarily on individual category names or a limited number of textual descriptions. Such approaches may fail to fully exploit the semantic diversity offered by alternative linguistic formulations.

Information fusion offers a principled mechanism to address this limitation by combining complementary sources of evidence under uncertainty [6, 7, 8]. Instead of treating category prompts as isolated descriptors, semantically related prompts can be interpreted as multiple sources of evidence describing the same underlying object concept. Semantic variants, descriptive phrases, contextual descriptions, and lexical relationships can provide complementary information that improves semantic coverage and increases the likelihood of successful object recognition. At the same time, combining multiple sources of semantic evidence introduces challenges regarding redundancy, uncertainty, semantic conflicts, and score aggregation.

This study investigates the fusion of multi-source semantic evidence for detecting objects from the long-tail distribution using OWL-ViT. The overall workflow of the proposed framework is illustrated in Figure 1. For each category, multiple semantic prompt sources are generated using lexical and descriptive semantic relationships. These prompts are evaluated independently and subsequently aggregated via alternative fusion mechanisms. The framework also examines uncertainty characteristics, semantic complementarity, the integration of long-tail priors, and framework stability across two widely used object detection datasets.

1.1. Gap Analysis

Existing prompt-based object detection approaches typically employ a single category prompt or a limited set of textual descriptions for each object category [3, 9, 4, 10]. Consequently, limited attention has been paid to understanding how multiple semantically related prompts interact as complementary sources of evidence within a unified information fusion framework—a phenomenon observed in several open-vocabulary detection frameworks [3, 9, 4, 10]. Although such methods provide semantic guidance, they do not explicitly model semantic evidence as a multi-source information fusion problem. Thus, limited attention has been devoted to understanding how multiple semantically related prompts interact, how semantic redundancy and complementarity influence detection results, and how long-tail category characteristics affect semantic evidence aggregation.

Furthermore, previous studies often focus on improving recognition performance without examining behavior under uncertainty, patterns of seman-

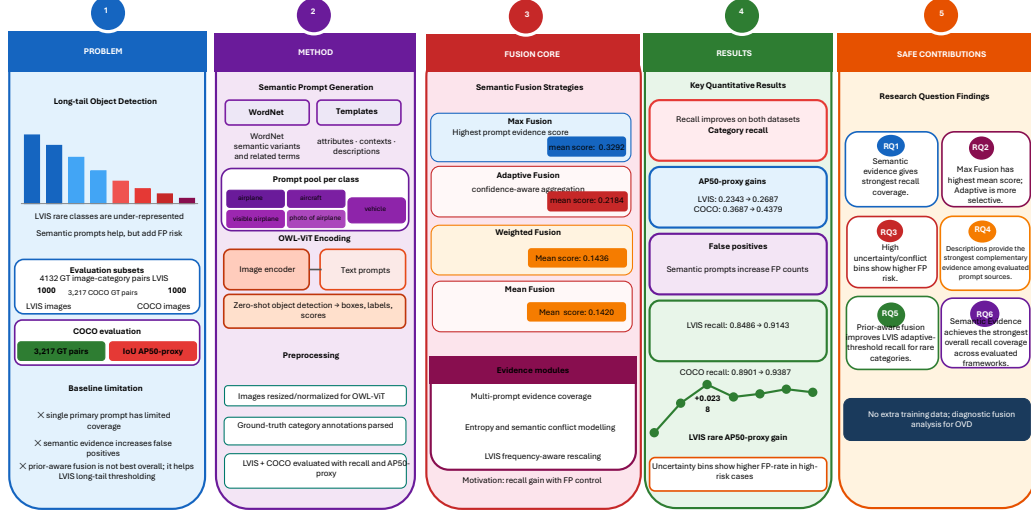


Figure 1: An overview of the long-tail challenge in open-vocabulary object detection. The left panel displays the category frequency distribution of the LVIS v1.0 dataset, highlighting the significant disparity between frequent and rare object categories. While frequent categories are represented by thousands of training examples, 418 rare categories contain fewer than ten training images. The right panel illustrates the limitations of single-prompt object detection under long-tail conditions, where insufficient semantic coverage can hinder the detection of rare and semantically diverse categories. This challenge motivates the proposed framework for fusing semantic information from multiple sources, which combines complementary semantic prompts to enhance category coverage and detection robustness.

62 tic conflict, or the effects of category frequency. These factors are particularly
63 important for the recognition of rare objects, as rare categories may bene-
64 fit from semantic enrichment differently than frequent categories. The lack
65 of systematic analyses regarding the fusion of semantic cues motivates the
66 present study.

67 1.2. Research Questions

68 The aim of this study is to investigate the fusion of semantic cues for rare
69 object recognition using vision-language models and principles of informa-
70 tion fusion. To systematically evaluate the proposed framework, six research
71 questions are formulated.

72 1. RQ1: Semantic Evidence Fusion

- 73 How can multiple semantically related textual prompts be fused as
 74 correlated semantic evidence sources to improve robust detection of
 75 long-tail object categories?
- 76 **2. RQ2: Fusion Strategy**
 77 Which semantic evidence aggregation strategy provides the most effective
 78 balance between semantic coverage and detection performance?
- 79 **3. RQ3: Uncertainty and Conflict Modelling**
 80 How do uncertainty and semantic conflict characteristics influence false-
 81 positive behaviour within semantic evidence fusion systems?
- 82 **4. RQ4: Redundancy versus Complementarity**
 83 To what extent do semantic redundancy and semantic complementarity
 84 contribute to category-level detection performance?
- 85 **5. RQ5: Prior-Aware Semantic Fusion**
 86 Can category-frequency prior information improve semantic evidence
 87 fusion for long-tail object categories?
- 88 **6. RQ6: Framework Stability and Evolution**
 89 How stable is the complete semantic evidence fusion framework across
 90 datasets, category-frequency groups, and fusion mechanisms?

91 1.3. Problem Statement

92 Long-tail object detection remains one of the major challenges in modern
 93 computer vision because object categories in real-world datasets are not uni-
 94 formly distributed. Frequent categories are represented by a large number of
 95 observations, whereas rare categories often provide only limited semantic and
 96 visual cues. Consequently, rare categories frequently exhibit lower detection
 97 rates and unstable detection behavior.

98 Recent vision-language models, such as OWL-ViT, have demonstrated the
 99 ability to perform object detection using text descriptions as semantic guid-
 100 ance. However, most existing approaches rely on a single category prompt
 101 or a limited set of manually defined prompts. Such approaches fail to fully
 102 exploit the rich semantic information available through alternative linguis-
 103 tic descriptions, lexical relationships, contextual descriptions, and category
 104 specific semantic variants.

105 Furthermore, existing studies rarely analyze the fusion of semantic evi-
 106 dence from the perspective of information fusion. Issues regarding semantic
 107 redundancy, semantic complementarity, uncertainty behavior, semantic con-
 108 flicts, and the effects of category frequency remain insufficiently explored.

109 Consequently, there is limited understanding of how multiple sources of se-
110 mantic evidence interact and how these interactions influence performance
111 in rare object detection.

112 The central problem addressed in this study is, therefore, the development
113 of a framework for fusing semantic evidence that is capable of generating,
114 aggregating, and analyzing multiple semantically related prompts to improve
115 semantic coverage and support robust rare object detection using vision-
116 language models.

117 1.4. *Novelty of this study*

118 The novelty of this study lies in formulating semantic prompt aggregation
119 as an information fusion problem for long-tail object recognition. Instead of
120 treating category prompts as isolated descriptors, the proposed framework
121 models multiple semantically related prompts as correlated sources of seman-
122 tic evidence that collectively contribute to object category recognition.

123 The proposed framework integrates primary category prompts, semantic
124 variants derived from WordNet, template prompts, and descriptive semantic
125 prompts into a unified semantic evidence generation process [11, 1, 3]. Subse-
126 quently, several aggregation mechanisms including mean fusion, max fusion,
127 weighted fusion, adaptive fusion, and prior-aware fusion—are evaluated to
128 analyze how different evidence combination strategies influence recognition
129 performance.

130 Another innovation is the comprehensive examination of semantic evi-
131 dence characteristics that goes beyond conventional detection results. The
132 study analyzes behavior under uncertainty, patterns of semantic conflict, re-
133 lationships of redundancy and complementarity, category frequency effects,
134 and the framework’s stability across LVIS and COCO evaluation subsets.
135 This combination of semantic evidence generation, evidence aggregation, un-
136 certainty analysis, and the integration of long-tail prior knowledge offers a
137 comprehensive perspective on information fusion for object-based detection
138 in vision-language systems.

139 1.5. *Significance of Our Work*

140 The significance of this work stems from the growing importance of se-
141 mantic logic in modern object detection systems. Although vision-language
142 models have significantly improved semantic understanding in computer vi-
143 sion, long-tail object categories continue to pose major challenges due to
144 limitations in semantic and visual representations. Consequently, there is

145 an increasing need for approaches that can improve semantic coverage and
146 enhance the robustness of object detection systems in the face of category
147 imbalances.

148 From an information fusion perspective, this study demonstrates how
149 multiple semantically related inputs can be integrated as complementary
150 sources of evidence, rather than being treated as isolated category descrip-
151 tions. By examining semantic evidence generation, semantic evidence aggre-
152 gation, behavior under uncertainty, semantic conflict patterns, and the effects
153 of category frequency, the proposed framework offers a structured methodol-
154 ogy for analyzing semantic information in vision and language based object
155 recognition systems.

156 The proposed framework contributes to research in information fusion
157 by introducing a practical workflow for aggregating semantic evidence and
158 investigating the interplay between semantic redundancy, semantic comple-
159 mentarity, uncertainty, and category frequency information. Furthermore,
160 the study contributes to computer vision research by providing empirical ev-
161 idence regarding the effectiveness of diversifying semantic inputs and fusing
162 semantic evidence for long-tail objects.

163 The results of this work could support future research into semantic rea-
164 soning, multimodal information integration, prompt engineering, uncertainty
165 aware object detection, and the analysis of vision-language models. Further-
166 more, the proposed methodology offers a reproducible framework that can
167 be extended to other problems of semantic evidence fusion beyond object
168 detection.

169 2. Literature Review

170 Object detection has made significant progress driven by deep learning,
171 transformer architectures, and vision-language models. Recent developments
172 have extended traditional object detection systems beyond closed-set classi-
173 fication and enabled semantic reasoning through natural language descrip-
174 tions. At the same time, long-tailed category distributions, semantic un-
175 certainty, and information integration remain major challenges. This sec-
176 tion covers the most relevant research areas related to the proposed study,
177 including open-vocabulary object detection, long-tail learning, multi-source
178 information fusion, and uncertainty quantification. The literature reviewed
179 establishes the theoretical foundation for the proposed semantic evidence
180 fusion framework and highlights the research gaps addressed by this study.

2.1. Open-Vocabulary Object Detection Using Vision-Language Models

Traditional object detection systems operate under a closed-set assumption, requiring all object categories to be predefined during training. Although such approaches have achieved remarkable performance gains, they encounter limitations when faced with novel or semantically diverse object categories. With the advent of vision-language models, a new paradigm has emerged that integrates textual descriptions and visual representations within a unified semantic space.

CLIP demonstrated that large-scale image-text pre-training enables strong semantic alignment between visual and textual modalities and supports zero-shot visual recognition via natural language supervision [1]. By learning generalized image-text representations from large-scale web data, CLIP laid the foundation for numerous tasks in open-vocabulary vision. Subsequent research extended these ideas to object detection by integrating textual inputs into detection frameworks and enabling category-level visual grounding through semantic descriptions.

OWL-ViT represents one of the most influential architectures for vision-language object detection and is based on the Vision Transformer (ViT), which demonstrated the effectiveness of Transformer architectures for the large-scale learning of visual representations [12]. The framework reformulates object detection as a vision-language matching problem, enabling detection via textual category descriptions without the need for task specific retraining. By leveraging Transformer-based image and text encoders, OWL-ViT offers a flexible mechanism for prompt driven object detection and supports semantic exploration beyond conventional detector architectures.

Several studies have investigated prompt engineering strategies to enhance the performance of vision-language models. Category names, template prompts, contextual descriptions, and semantic reformulations have all been explored as mechanisms to increase semantic alignment between text representations and visual content. However, most existing approaches rely on a limited number of prompts and typically evaluate them independently. Consequently, relatively little attention has been paid to treating multiple semantic prompts as complementary sources of information within a unified information fusion framework.

Despite significant progress in open-vocabulary detection, challenges remain regarding semantic coverage, sensitivity to prompts, behavior under uncertainty, and performance on rare categories. Rare object categories often possess limited semantic representation and could benefit from the generation

219 of richer semantic evidence. These limitations motivate the development of
220 semantic evidence fusion approaches capable of integrating multiple semanti-
221 cally related prompts to improve category coverage and detection robustness.

222 Several recent open-vocabulary detection frameworks have extended vi-
223 sion language pre-training to object localization including ViLD, Detic,
224 GLIP, RegionCLIP, GroundingDINO, OV-DETR, and OWL-ViT [13, 9, 4,
225 10, 5, 14, 3]. While these approaches demonstrate the effectiveness of seman-
226 tic supervision, they generally evaluate a limited number of prompts and do
227 not explicitly model semantic evidence fusion.

228 Furthermore, recent studies have shown that large-scale vision-language
229 pre-training can improve semantic generalization beyond fixed category vo-
230 cabularies, motivating the use of richer semantic descriptions during object
231 detection [1, 15].

232 2.2. Long-Tail Learning and Class Re-balancing

233 Long-tail learning has emerged as an important area of research because
234 real-world datasets rarely exhibit balanced category distributions. Instead,
235 a few frequent categories typically dominate the majority of observations,
236 while many categories occur only rarely. This imbalance poses significant
237 challenges for recognition and detection systems, as learning algorithms are
238 often biased in favor of dominant categories, leaving rare classes underrepre-
239 sented.

240 The LVIS dataset was specifically introduced to evaluate object recogni-
241 tion under long-tail category distributions and to highlight the limitations of
242 conventional detection systems in handling rare categories [16]. The dataset
243 categorizes object classes into rare, frequent, and very frequent groups, pro-
244 viding a practical benchmark for analyzing category frequency effects. Nu-
245 merous studies have shown that detection performance for rare categories is
246 often limited due to semantic and visual representation.

247 Traditional long-tail learning approaches have primarily focused on class
248 re-weighting strategies. These include resampling techniques, cost sensitive
249 learning, variants of focal loss, class-balanced loss functions, and decoupled
250 representation learning. Such approaches attempt to mitigate category im-
251 balance by altering training distributions or adjusting optimization objec-
252 tives. While these methods can improve performance on rare categories, they
253 generally require retraining and often lead to trade-offs between accuracy for
254 rare and frequent categories.

Recent research has explored semantic approaches to address long-tail challenges. Vision-language models provide access to semantic knowledge embedded in textual representations, thereby offering an alternative mechanism to support the recognition of rare categories. Instead of relying solely on visual observations, semantic descriptions can provide complementary cues that enrich category representations. However, most existing studies examine semantic cues in isolation and do not systematically analyze how multiple semantic descriptions can be integrated for rare object detection.

Consequently, there remains a need for methods that leverage semantic diversity while preserving the flexibility of vision-language architectures. The present study addresses this issue by investigating semantic evidence fusion and category frequency-aware semantic aggregation in the context of rare object detection.

Recent studies have shown that long-tail recognition remains one of the major challenges in modern computer vision, as model performance is often biased toward frequent categories while rare classes are underrepresented [17]. Existing solutions include class-balanced learning, re-sampling, re-weighting, and specialized loss functions such as Seesaw Loss [18].

Furthermore, large-scale analyses of real-world recognition systems demonstrate that the long-tail imbalance continues to pose a significant obstacle to robust visual recognition, given that rare categories receive substantially less supervision than dominant classes [19].

Recent surveys indicate that long-tail recognition remains a persistent challenge in visual recognition tasks, despite significant advances in representation learning and class re-balancing methods. [20].

2.3. Multi-Source Information Fusion and Evidence Theory

Information fusion deals with the integration of multiple information sources to improve decision quality and robustness. Established literature has identified information fusion as a fundamental framework for combining heterogeneous evidence under uncertainty [21]. Information fusion techniques have been successfully applied in fields such as sensor networks, robotics, surveillance systems, medical diagnostics, autonomous vehicles, and decision support systems. The fundamental principle behind information fusion is that different information sources can provide complementary evidence regarding the same underlying phenomenon.

Several information fusion frameworks have been proposed in the literature. Bayesian inference methods combine probabilistic evidence under un-

292 certainty, while Dempster Shafer theory offers a mechanism for representing
293 and combining uncertain beliefs from multiple sources of evidence. Similarly,
294 ensemble learning methods aggregate predictions from multiple models to
295 improve robustness and generalization performance.

296 In the field of computer vision, information fusion has traditionally fo-
297 cused on combining multiple sensors, modalities, or model predictions. Ex-
298 amples include RGB-depth fusion, image-text fusion, multi-modal scene un-
299 derstanding, and ensemble object detection systems. However, comparatively
300 little research has addressed semantic prompt fusion as an information fusion
301 problem.

302 Recent advances in vision-language models have opened up new possibil-
303 ities for semantic fusion, as text prompts can be interpreted as independent
304 sources of semantic cues. Different prompt formulations can emphasize var-
305 ious aspects of the same object concept, thereby providing complementary
306 information. At the same time, however, semantic prompts can introduce
307 redundancy, uncertainty, and conflicts. Understanding the nature of these
308 interactions is crucial for developing effective frameworks for fusing semantic
309 cues.

310 The proposed study adopts this information fusion approach by treating
311 semantically related prompts those describing the same object category as
312 correlated sources of semantic cues. Multiple types of semantic prompts are
313 generated and subsequently combined using various fusion mechanisms to
314 analyze how the integration of semantic cues affects object detection perfor-
315 mance.

316 Research in information fusion has yielded several fundamental frame-
317 works for evidence aggregation under uncertainty, including Bayesian infer-
318 ence, Dempster Shafer theory, formulations of belief entropy, and conflict
319 aware evidence combination methods [22, 23, 24, 8, 7]. These approaches pro-
320 vide a theoretical basis for treating semantic prompts as correlated sources
321 of semantic evidence.

322 Beyond evidence theory, related concepts have also been investigated in
323 ensemble learning and multi-view learning, where complementary informa-
324 tion sources are combined to enhance robustness and predictive performance
325 [25, 26].

326 Alternative representations of uncertainty, such as fuzzy set theory, have
327 also been widely used for reasoning with imprecise information [27].

2.4. *Uncertainty Quantification and Calibration in Deep Learning*

Despite the remarkable success of deep learning models, prediction uncertainty remains a significant challenge in practical machine learning systems. Modern neural networks often produce highly confident predictions, even when operating under ambiguous, noisy, or previously unseen conditions. Consequently, understanding and quantifying uncertainty has become a major area of research in computer vision and artificial intelligence.

Uncertainty estimation aims to measure the reliability of model predictions and provide additional information regarding confidence and decision quality. Uncertainty plays a crucial role in object detection, as detection systems must simultaneously determine an object’s location, category, and confidence scores. Inaccurate confidence estimation can lead to increased false alarms, reduced decision reliability, and adverse effects on downstream applications.

Several approaches to uncertainty quantification in deep learning have been proposed. Bayesian neural networks model uncertainty via probability distributions over model parameters, whereas Monte Carlo dropout approximates Bayesian inference through stochastic forward passes. Ensemble based approaches estimate uncertainty by analyzing the variability of predictions across multiple models. More recently, uncertainty estimation methods have been integrated into object detection frameworks to improve prediction reliability and calibration.

Prediction calibration represents a closely related research topic. Recent surveys on uncertainty quantification have emphasized that reliable uncertainty estimates are essential for safety critical machine learning applications, particularly when decisions are made based on incomplete or ambiguous evidence [28]. Furthermore, calibration studies show that modern deep neural networks often yield overconfident predictions despite high predictive accuracy [29].

A well calibrated model provides confidence estimates that accurately reflect the probability of correct predictions. Calibration metrics such as Expected Calibration Error (ECE), reliability diagrams, and confidence-to-error ratios are frequently used to evaluate prediction quality. Although calibration has received significant attention in classification tasks, calibration and uncertainty behavior remain active areas of research in object detection and vision-language systems.

Vision-language models introduce additional sources of uncertainty because text prompts influence semantic interpretation during inference. Dif-

366 ferent prompt formulations can yield varying confidence levels and provide
367 potentially conflicting semantic cues. Consequently, analyzing uncertainty
368 behavior becomes particularly important when combining multiple sources
369 of semantic information. Semantic conflicts between prompts may indicate
370 ambiguity, disagreements, or reduced reliability in evidence integration.

371 This study incorporates uncertainty and semantic conflict analyses as
372 integral components of the proposed framework. Rather than merely evalu-
373 ating detection results, the framework examines how uncertainty character-
374 istics relate to false-positive behavior and the quality of semantic evidence.
375 This analysis provides additional insights into the reliability and robustness
376 of semantic evidence fusion in long-tail object detection systems.

377 Prompt generation was inspired by recent studies on prompt learning
378 and semantic descriptions, which demonstrate that alternative linguistic for-
379 mulations can enhance semantic alignment within vision-language models
380 [30, 31, 32, 33]. Additionally, WordNet semantic relations were utilized to
381 automatically generate synonym and hypernym variants [11].

382 Deep ensembles have also established themselves as a practical and effec-
383 tive approach to predictive uncertainty estimation in deep neural networks
384 [34].

385 Recent surveys have highlighted uncertainty estimation as a key prereq-
386 uisite for trustworthy artificial intelligence systems operating under distribu-
387 tion shift and incomplete evidence [35].

388 2.5. Literature Summary

389 The literature discussed in this section highlights significant advances
390 in vision-language models, rare object detection, information fusion, and
391 uncertainty quantification. Vision-language architectures such as OWL-ViT
392 have demonstrated the effectiveness of semantic guidance in object detection,
393 while research on long-tail learning has underscored the persistent challenges
394 associated with rare object categories. Research into information fusion has
395 shown that integrating multiple sources of evidence can improve decision
396 quality, and studies analyzing semantic disagreements have highlighted the
397 importance of reliability analysis in modern machine learning systems.

398 Despite these advances, several limitations remain evident. Existing
399 prompt-based object detection approaches often rely on single category
400 prompts or a limited number of text descriptions. Research on long-tail
401 object detection focuses primarily on training based balancing strategies and
402 rarely explores the diversity of semantic cues as a complementary source of

information. Furthermore, relatively little attention has been paid to treating multiple semantic prompts as correlated sources of semantic cues within an information fusion framework. The interplay between semantic redundancy, semantic complementarity, uncertainty behavior, semantic conflict, and category frequency information remains insufficiently explored.

These observations reveal a clear research opportunity to investigate semantic evidence fusion for the detection of rare objects. The proposed study addresses this gap by developing a framework for multi-source semantic evidence that integrates semantic cue generation, evidence aggregation, uncertainty analysis, complementarity assessment, and frequency-aware prior modeling. Table 1 summarizes the research gaps motivating the proposed framework. The following section presents the methodology and implementation details of the proposed framework.

3. Methodology

This section presents the methodology used to investigate semantic evidence fusion in long-tail object detection. The proposed framework integrates semantic prompt generation, semantic evidence aggregation, uncertainty analysis, redundancy complementarity assessment, and frequency-aware pre-modeling within a unified vision-language object detection pipeline. The framework is implemented using OWL-ViT and evaluated on curated subsets of the LVIS and COCO datasets. The overall workflow of the proposed methodology is illustrated in Figure 2.

WordNet prompt variants were automatically generated using synonym and hypernym relationships available via the NLTK WordNet interface [11]. The use of lexical semantic relationships is supported by classical research on semantic similarity, which has demonstrated that hierarchical semantic structures provide meaningful representations of conceptual relationships [38].

3.1. Dataset

The experimental evaluation was conducted using subsets of the LVIS and COCO datasets, which are widely used benchmarks in object detection research. These datasets offer diverse object categories and complex visual scenes, enabling the evaluation of semantic evidence fusion across various category distributions.

The LVIS dataset was selected because it explicitly features long-tail category distributions and provides frequency annotations that categorize ob-

Table 1: A literature review and analysis of research gaps motivate the proposed framework for semantic evidence fusion. Existing studies have independently advanced open-vocabulary detection, long-tail detection, prompt learning, uncertainty estimation, and information fusion. However, no prior work has collectively investigated semantic prompt diversity, semantic evidence fusion, uncertainty-aware reasoning, redundancy analysis, and frequency-aware semantic calibration within a unified long-tail, open-vocabulary object detection framework.

Research Area	Representative Studies	Key Contributions	Remaining Limitation and Motivation for This Work
Vision-Language Representation Learning	CLIP [1], ALIGN [36], Florence [2]	Established transferable vision-language representations capable of supporting zero-shot visual recognition and semantic generalisation beyond fixed category vocabularies.	Focus primarily on representation learning rather than object-level semantic evidence aggregation. Multiple semantic descriptions are not explicitly combined during inference.
Open-Vocabulary Object Detection	OWL-ViT [3], ViLD [13], Detic [9], GroundingDINO [5], RegionCLIP [10]	Extended vision-language representations to object detection and enabled recognition of categories beyond supervised training labels.	Most approaches rely on individual category prompts or limited textual descriptions and do not investigate semantic evidence fusion across multiple prompt sources.
Prompt Learning and Prompt Engineering	CoOp [30], CoCoOp [31], MaPLe [32]	Demonstrated that prompt design strongly influences vision-language model performance and semantic alignment.	Prompt optimisation is typically performed using a single learned prompt representation and does not analyse complementary semantic information from multiple prompt categories.
Long-Tail Recognition	LVIS [16], Seesaw Loss [18], Decoupled Long-Tail Learning [37]	Addressed severe class imbalance and improved recognition performance for under-represented categories.	Methods remain largely visual-feature driven and do not exploit semantic prompt diversity as an additional source of information for rare-category recognition.
Information Fusion and Evidence Combination	Dempster-Shafer Theory [23], Khaleghi et al. [6], Denoeux [7]	Provided principled frameworks for combining heterogeneous evidence under uncertainty and conflict.	Existing fusion frameworks have not been adapted to semantic prompt reasoning within open-vocabulary object detection systems.
Uncertainty Quantification	Guo et al. [29], Abdar et al. [28], Lakshminarayanan et al. [34]	Developed techniques for confidence calibration, uncertainty estimation, and reliability assessment in machine learning systems.	Current approaches do not explicitly model semantic conflicts among multiple prompt-derived evidence sources in long-tail open-vocabulary detection.
Semantic Knowledge Integration	WordNet [11], Description-based prompting [33]	Demonstrated the usefulness of lexical relationships and semantic descriptions for improving concept representations.	Semantic relations are generally used independently and are not systematically fused, weighted, and evaluated within an evidence-based detection framework.
Proposed Semantic Evidence Fusion Framework	This Work	Introduces semantic prompt expansion, semantic evidence fusion, fusion-strategy evaluation, uncertainty and conflict modelling, redundancy analysis, prior-aware semantic weighting, and framework-level evaluation using LVIS and COCO benchmarks.	Provides a unified Information Fusion framework for long-tail open-vocabulary object detection that jointly addresses semantic diversity, evidence aggregation, uncertainty reasoning, and category-frequency awareness.

438 ject classes into rare, common, and frequent groups [16]. This characteristic
 439 makes LVIS particularly suitable for analyzing category-frequency effects and
 440 evaluating frequency-aware semantic fusion strategies. The COCO dataset
 441 was included to provide a complementary evaluation environment featuring a
 442 different category distribution and object composition. The Microsoft COCO
 443 benchmark remains one of the most widely used datasets for object detec-
 444 tion and scene understanding, owing to its large-scale object annotations and
 445 diverse visual content [39].

446 To ensure computational consistency and facilitate controlled experi-
 447 ments, curated subsets were constructed from both datasets. A total of

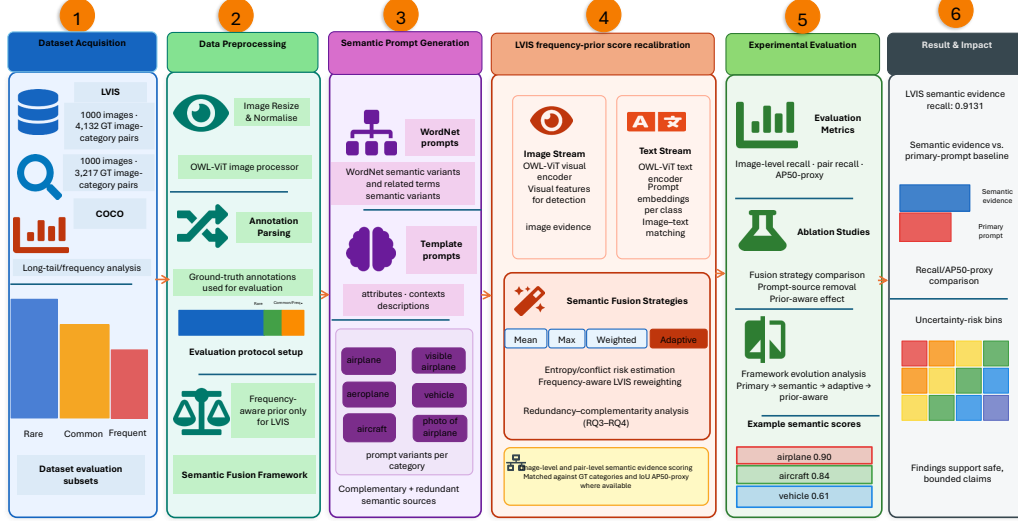


Figure 2: Overall methodological workflow of the proposed multi-source semantic evidence fusion framework. The pipeline comprises dataset preparation, semantic prompt generation, OWL-ViT based evidence extraction, semantic evidence aggregation, and frequency stratified evaluation.

1,000 images from LVIS and 1,000 images from COCO were evaluated. These subsets offer sufficient category diversity while maintaining manageable computational requirements for extensive semantic prompt evaluation and fusion analysis.

An overview of the datasets used in this study is shown in Figure 3. Ground truth annotations were used exclusively for evaluation and performance measurement. The generation of semantic prompts and object detection inference were performed independently of the ground-truth categories to ensure that the generation of semantic cues remained separate from the evaluation process.

The use of two datasets enables the analysis of the proposed framework across varying object distributions, category frequencies, and semantic characteristics. This dual dataset design also facilitates an examination of the framework’s stability and its generalization behavior across multiple evaluation scenarios.

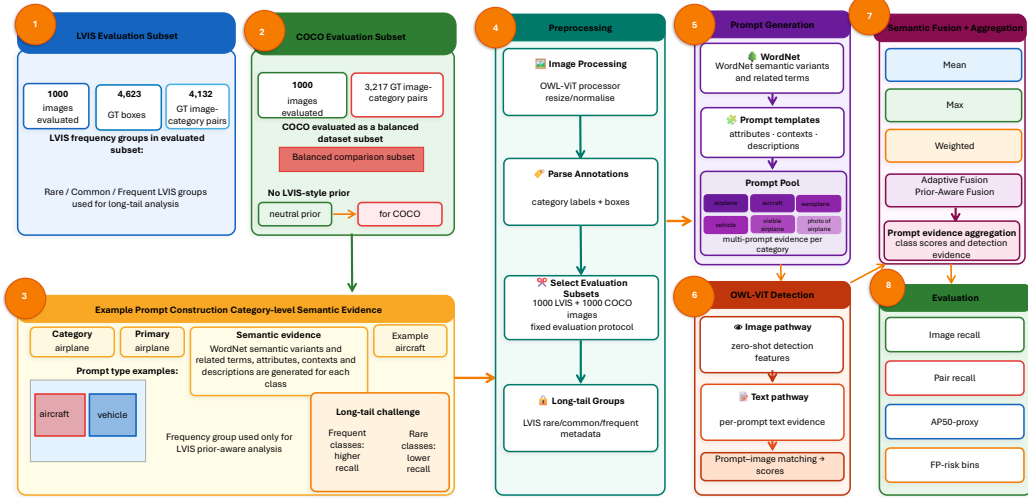


Figure 3: Overview of the LVIS and COCO evaluation subsets used in this study.

3.2. Detailed Methodology

The proposed framework formulates semantic prompt aggregation as a multi-source information fusion problem for rare object detection. Instead of relying on a single textual category description, multiple semantically related prompts are generated and treated as correlated sources of semantic evidence. These sources of evidence are subsequently aggregated using various fusion mechanisms to investigate their contribution to object detection performance.

The overall approach comprises five main phases: semantic prompt generation, semantic evidence detection, semantic evidence fusion, uncertainty and conflict analysis, and framework level evaluation. The complete workflow is illustrated in Figure 2.

The object detection backbone used in this study is OWL-ViT, a transformer-based vision architecture specifically, a vision-language architecture developed for prompt-based object detection [3]. OWL-ViT projects textual descriptions and image regions into a shared semantic embedding space and performs object detection via semantic similarity matching. Unlike conventional object detectors, OWL-ViT allows object categories to be specified via text inputs, thereby providing a suitable platform for investigating semantic evidence generation and aggregation.

482 For each object category, an initial primary prompt consisting of the
483 category name itself is generated. Additional semantic evidence is subse-
484 quently constructed using multiple prompt-generation mechanisms. Word-
485 Net lexical relationships are utilized to obtain semantically related category
486 descriptions—including synonym and hypernym variations, where available
487 [11]. Furthermore, contextual prompt templates are employed to generate al-
488 ternative linguistic phrasings, while descriptive prompt construction provides
489 richer semantic descriptions intended to enhance semantic coverage.

490 Each generated prompt is independently evaluated by OWL-ViT. For
491 every image, the detector produces bounding-box predictions and associated
492 confidence scores for all active prompts. These outputs constitute the seman-
493 tic evidence pool from which subsequent fusion operations are performed.
494 Because each prompt may capture different semantic characteristics of an
495 object category, the resulting evidence sources may exhibit both redundancy
496 and complementarity.

497 Several fusion strategies are investigated to aggregate semantic evi-
498 dence. Mean fusion calculates the average evidence value across all semantic
499 prompts, providing a simple basis for aggregation. Max fusion selects the
500 strongest observed semantic evidence value across the prompts, thereby fo-
501 cusing on high-confidence matches. Weighted fusion combines evidence using
502 weighted scores to balance the contributions of individual prompt sources.
503 Adaptive fusion dynamically adjusts evidence aggregation based on the char-
504 acteristics of the semantic evidence, while prior-aware fusion incorporates
505 category frequency information derived from long-tail category groups within
506 LVIS.

507 Following semantic aggregation, measures of uncertainty and semantic
508 conflict are calculated. Uncertainty analysis examines the variability of evi-
509 dence values generated by different semantic prompts. Significant differences
510 between prompts may indicate ambiguity or reduced semantic reliability. The
511 analysis of semantic conflicts quantifies the extent to which different prompt
512 sources provide contradictory cues for the same object category. These mea-
513 surements provide additional diagnostic information regarding the reliability
514 of the fusion of semantic cues.

515 The framework also evaluates redundancy and complementarity among
516 the semantic prompt sources. Prompt ablation analysis is conducted by
517 systematically removing individual prompt categories and measuring the re-
518 sulting impact on semantic cue coverage. This procedure makes it possible
519 to identify the prompt types that make the strongest complementary contri-

520 bution to object detection.

521 Frequency-aware prior modeling is incorporated to examine the behavior
 522 of long-tail categories. The categories are grouped according to LVIS fre-
 523 quency annotations, and category-frequency priors are integrated into the
 524 fusion process. This mechanism enables the aggregation of semantic evi-
 525 dence to account for differences between rare, frequent, and very frequent
 526 categories, offering an additional perspective on the fusion of long-tail se-
 527 mantic evidence.

528 Finally, the complete framework is evaluated based on six research ques-
 529 tions addressing the fusion of semantic evidence, the behavior of the fusion
 530 strategy, the modeling of uncertainty and conflict, redundancy complemen-
 531 tarity relationships, prior-aware fusion, and the stability of the framework.
 532 The resulting analyses offer a comprehensive information-fusion perspective
 533 on semantic prompt aggregation for rare object detection. All AP50 values
 534 reported in this section correspond to the AP50 proxy metric defined in the
 535 evaluation criteria section and are used exclusively for the relative compari-
 536 son of framework variants.

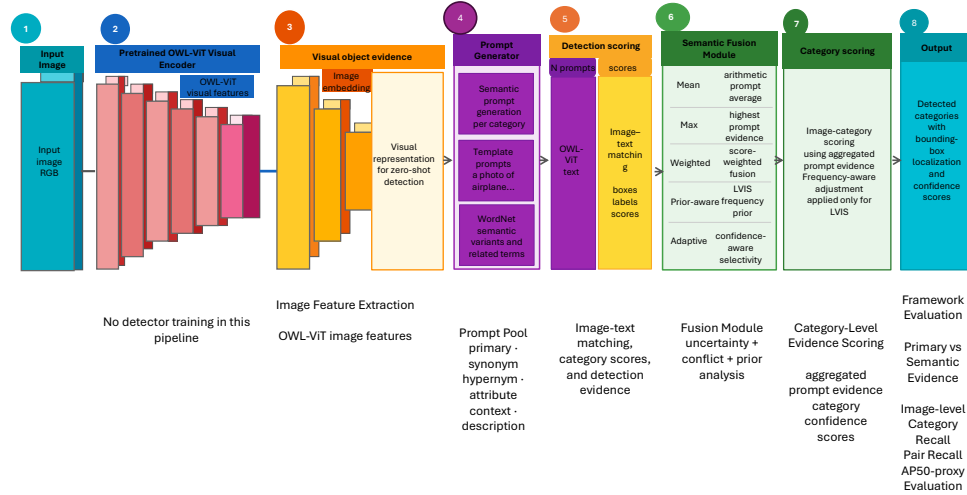


Figure 4: Architecture of the proposed framework for semantic evidence fusion. OWL-ViT is used to generate visual and semantic evidence based on prompts; this evidence is subsequently combined using multiple fusion strategies and evaluated through uncertainty, conflict, and priority-aware analysis before category-level assessment takes place.

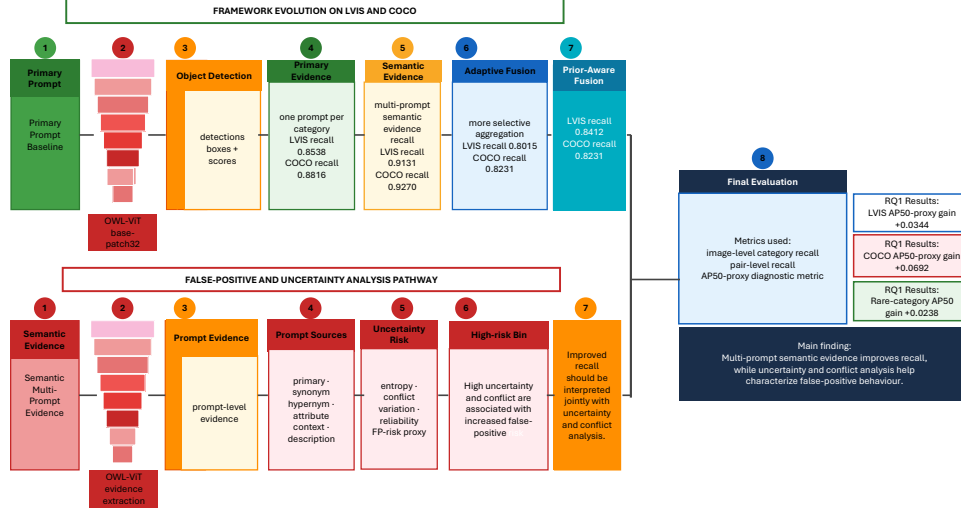


Figure 5: Summary of the framework’s development and the results of the uncertainty analysis. The figure illustrates the progression from primary cue detection through semantic evidence fusion and adaptive fusion to prior-aware fusion, as well as the uncertainty analysis approach used to characterize false alarm behavior and the reliability of semantic evidence.

3.3. Evaluation Metrics

The proposed framework was evaluated using several complementary metrics designed to assess the effectiveness of semantic evidence, fusion behavior, uncertainty characteristics, and framework stability. Since the study focuses on the fusion of semantic evidence rather than benchmark optimization, the evaluation centers on semantic coverage, image category detection behavior, and the performance of evidence aggregation.

Metrics based primarily on recall were used to assess the effectiveness of semantic evidence. Image category recall measures the proportion of ground truth image-category pairs successfully detected by the framework, providing a direct measure of semantic coverage. Additionally, category level recall was used to evaluate the ability of the semantic evidence fusion process to improve object detection compared to the baseline prompts.

To evaluate localization performance, AP50 proxy metrics were calculated using an Intersection-over-Union (IoU) threshold of 0.50. The AP50 proxy measure provides a consistent localization-based evaluation within the experi-

553 mental framework and enables comparative analysis across fusion strategies.
554 As the study focuses on controlled evaluation subsets rather than official
555 benchmark submissions, AP50 proxy values are interpreted as relative in-
556 dicators of localization effectiveness rather than as official COCO or LVIS
557 benchmark scores.

558 The quality of semantic evidence was analyzed using prompt-level and
559 fusion-level statistics. Mean semantic values, maximum semantic values,
560 prompt hit counts, and measures of semantic evidence coverage were cal-
561 culated to examine the contribution of various prompt sources. These met-
562 rics provide insight into how semantic evidence accumulates across multiple
563 prompt formulations.

564 Metrics for uncertainty and semantic conflict were incorporated to eval-
565 uate the reliability of evidence. Uncertainty metrics quantify the variabil-
566 ity among semantic sources of evidence, while conflict metrics estimate the
567 discrepancies between semantic inputs describing the same object category.
568 Higher uncertainty or conflict values may indicate ambiguous semantic evi-
569 dence and are therefore analyzed in the context of false alarms.

570 An analysis of redundancy and complementarity was conducted using
571 prompt elimination experiments. The contribution of individual prompt cat-
572 egories was measured by removing specific prompt sources and observing
573 the resulting change in semantic evidence coverage. Greater performance
574 degradation indicates a more significant complementary contribution from
575 the removed semantic source.

576 To analyze long-tail data, category-frequency-aware metrics were calcu-
577 lated using LVIS frequency annotations. Rare, frequent, and very frequent
578 categories were evaluated separately to investigate the impact of category
579 frequency information on the fusion of semantic evidence. Recall by fre-
580 quency group and AP50 proxy measures were used to assess the effectiveness
581 of prior-aware fusion in long-tail scenarios.

582 Finally, framework-level comparison metrics were employed to evaluate
583 performance evolution across primary prompt detection, semantic evidence
584 fusion, adaptive fusion, and prior-aware fusion. These comparisons provide a
585 comprehensive assessment of the framework’s stability and enable an analysis
586 of semantic evidence behavior across datasets and category frequency groups.

587 *3.4. Experimental Settings*

588 All experiments were conducted using the OWL-ViT vision-language ob-
589 ject detection framework, implemented within a Python-based evaluation

environment. Semantic prompt generation, semantic evidence fusion, uncertainty analysis, the assessment of redundancy and complementarity, and experiments on prior-aware fusion were carried out using a unified implementation pipeline to ensure consistency across all research questions.

The experimental evaluation utilized carefully selected subsets of the LVIS and COCO datasets, each comprising 1,000 images. These subsets were chosen to provide sufficient category diversity while maintaining computational feasibility for large-scale experiments regarding semantic prompt evaluation and evidence aggregation, as summarized in Table 2.

Table 2: Experimental and Network Configuration

Parameter	Configuration
Detection Framework	OWL-ViT (Vision-Language Object Detection)
Backbone Model	google/owlvit-base-patch32
Implementation Language	Python 3.x
Deep Learning Framework	PyTorch
Transformer Library	HuggingFace Transformers
Evaluation Datasets	LVIS v1.0 and MS COCO 2017
LVIS Evaluation Images	1000
COCO Evaluation Images	1000
Semantic Prompt Sources	Primary, Description, Contextual, Synonym, Hypernym, Attribute Prompts
Fusion Strategies	Mean, Max, Weighted, Adaptive, Prior-Aware Fusion
Primary Detection Threshold	0.05
Adaptive Threshold	0.10
IoU Threshold Used	0.50
Long-Tail Categories	Rare, Common, Frequent
Research Questions	RQ1–RQ6
Random Seed	42
Evaluation Mode	Inference Only (No Detector Retraining)

Semantic prompt generation utilized primary category prompts alongside semantically related prompt variants derived through lexical and descriptive semantic transformations. Several prompt categories were evaluated, includ-

602 ing primary prompts, descriptive prompts, contextual prompts, attribute
603 prompts, synonym prompts, and hypernym prompts, where applicable.

604 Object detection was performed independently for each semantic prompt
605 using OWL-ViT. The resulting semantic evidence outputs were combined us-
606 ing various fusion mechanisms, including mean fusion, max fusion, weighted
607 fusion, adaptive fusion, and priority-aware fusion. Consistent threshold set-
608 tings were maintained across all experiments to ensure a fair comparison
609 between fusion strategies and framework variants.

610 All research questions were evaluated using the same experimental pro-
611 tocol, enabling a direct comparison of the behavior of semantic evidence
612 across different datasets, category frequency groups, and fusion mechanisms.
613 The resulting framework provides a reproducible experimental environment
614 for analyzing the fusion of semantic evidence in long-tail object recognition
615 systems. These thresholds were selected empirically through preliminary ex-
616 periments and were kept constant across all evaluations.

617 4. Results

618 This section presents the experimental results obtained with the proposed
619 framework for semantic evidence fusion using the LVIS and COCO evaluation
620 subsets. The evaluation is organized according to the six research questions
621 introduced in Section 1.2. The experiments examine the performance of se-
622 mantic evidence fusion, strategies for aggregating semantic evidence, behav-
623 ior under uncertainty and conflict, redundancy and complementarity among
624 semantic prompt sources, category-frequency-aware fusion, and the overall
625 evolution of the framework. The reported results are presented in tables and
626 figures generated from the experimental pipeline, providing a quantitative
627 assessment of semantic evidence behavior across the various components of
628 the proposed framework.

629 4.1. Semantic Evidence Fusion Performance (RQ1)

630 This experiment investigates whether multiple semantically related
631 prompts can improve object detection compared to a single category prompt.
632 For each object category, a primary prompt was first evaluated using OWL-
633 ViT. Subsequently, additional semantic prompts were generated using se-
634 mantic expansion techniques, including descriptive and lexical variations.
635 The outputs from all prompts were then combined via a semantic evidence
636 fusion framework and evaluated on COCO and LVIS subsets. The goal was

637 to determine whether semantic evidence generated from multiple prompt
638 sources increases category coverage, improves recall, and yields more robust
639 detection performance than detection using only the primary prompt.

640 Table 3 presents a dataset-level comparison between Primary Prompt
641 Detection and Semantic Evidence Fusion. For the COCO subset, recall in-
642 creased from 0.8901 to 0.9387, representing an absolute gain of 0.0486 and a
643 relative improvement of 5.46%. AP50-proxy rose from 0.3687 to 0.4379, while
644 the number of detected image-category pairs increased from 22,093 to 35,020.
645 For the LVIS subset, recall increased from 0.8486 to 0.9143, representing an
646 absolute gain of 0.0657 and a relative improvement of 7.74%. AP50-proxy
647 rose from 0.2343 to 0.2687, while the number of detected image-category
648 pairs increased from 29,838 to 47,870. The results demonstrate that incorpo-
649 rating multiple semantic prompt sources enhanced both semantic coverage
650 and detection performance across the evaluated datasets.

Table 3: Dataset-level performance comparison between Primary Prompt Detection and Semantic Evidence Fusion on the COCO and LVIS evaluation subsets. Recall, AP50-proxy, detection statistics, and semantic evidence characteristics are reported.

Dataset	Images	Primary Recall	Semantic Recall	Abs. Gain	Rel. Gain (%)	Primary AP50	Semantic AP50	AP50 Gain	AP50 Rel. Gain (%)	Prim
COCO	1000	0.8901	0.9387	0.0486	5.46	0.3687	0.4379	0.0692	18.77	2
LVIS	1000	0.8486	0.9143	0.0657	7.74	0.2343	0.2687	0.0344	14.68	2

651 Table 4 presents the analysis of long-tail categories derived from the se-
652 mantic evidence fusion framework. Rare categories achieved a recall of 0.7674
653 and an improvement in the AP50 proxy from 0.0817 to 0.1055, representing
654 a relative increase of 29.13%. Frequent categories achieved a recall of 0.8722,
655 with AP50 proxy values of 0.1315 and 0.1372 for the Primary and Semantic
656 configurations, respectively. Very frequent categories achieved the highest
657 recall of 0.9298 and an increase in the AP50 proxy from 0.2615 to 0.2991.
658 The average number of prompt hits rose from 6.63 for rare categories to 12.03
659 for very frequent categories, indicating a greater accumulation of semantic
660 evidence for better-represented categories. Across all frequency groups, Se-
661 mantic Evidence Fusion yielded higher AP50 proxy values than the Primary
662 prompt configuration.

663 Figure 6 shows a comparison of recall values at the dataset level be-
664 tween primary prompt detection and semantic evidence fusion. The figure
665 visualizes the recall values achieved for the COCO and LVIS evaluation sub-
666 sets and illustrates the consistent improvement resulting from the inclusion
667 of multiple semantic prompt sources. The difference between the primary

Table 4: Long-tail category performance obtained using Semantic Evidence Fusion across the LVIS frequency groups. Recall, fusion statistics, prompt utilization, and AP50-proxy measurements are reported for rare, common, and frequent categories.

Group	Code	GT Pairs	Detected Pairs	Recall	Max Fusion	Adaptive Fusion	Prompt Hits	Primary AP50	Semantic AP50	AP50 Gain	AP50 Gain
Rare	r	331	254	0.7674	0.1797	0.1289	6.63	0.0817	0.1055	0.0238	29.1
Common	c	266	232	0.8722	0.2405	0.1637	11.88	0.1315	0.1372	0.0057	4.33
Frequent	f	3535	3287	0.9298	0.3302	0.2165	12.03	0.2615	0.2991	0.0376	14.3

and semantic configurations is evident in both datasets, with the greatest improvement observed in the LVIS subset. The figure demonstrates that aggregating semantic evidence increased the category-level detection rate across the evaluated datasets.

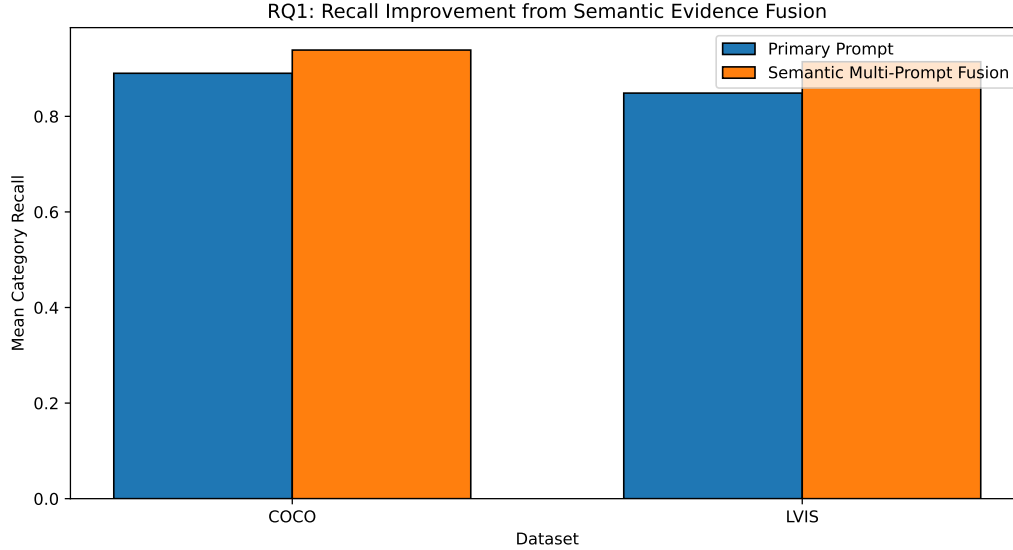


Figure 6: Dataset-level recall comparison between Primary Prompt Detection and Semantic Evidence Fusion for the COCO and LVIS evaluation subsets. The figure reports the recall values obtained using the final semantic evidence fusion framework.

Figure 7 shows a comparison of the AP50 proxy across LVIS frequency groups. The figure illustrates the performance differences between the "Primary Prompt" and "Semantic Evidence" configurations for rare, frequent, and very frequent categories. The greatest improvement in the AP50 proxy is observed in the rare category group, while smaller but consistent improvements are evident in the frequent and very frequent groups. The figure

678 demonstrates that the fusion of semantic evidence offers measurable benefits
 679 across the entire long-tail category structure and yields the strongest relative
 680 improvement for underrepresented categories.

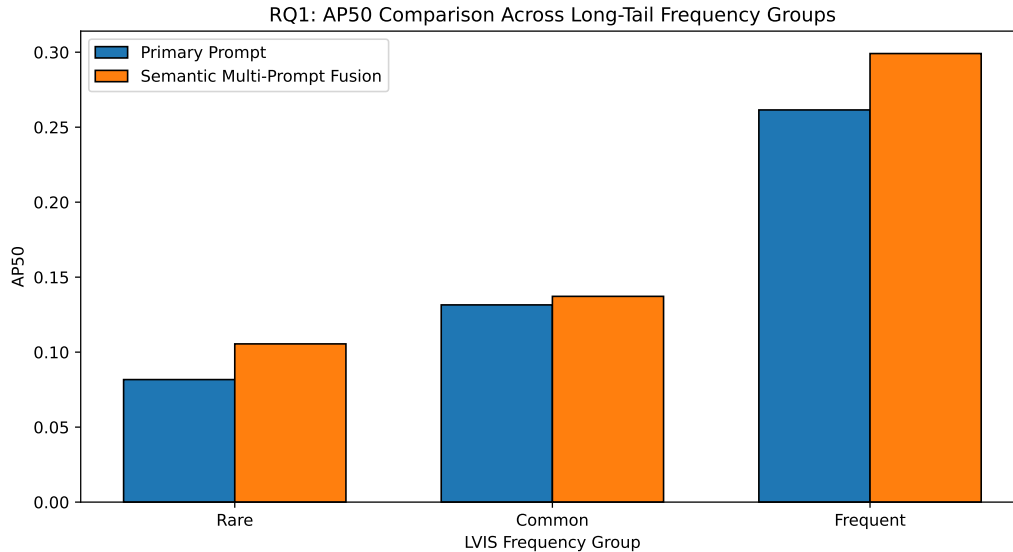


Figure 7: AP50-proxy comparison between Primary Prompt Detection and Semantic Evidence Fusion across the LVIS rare, common, and frequent category groups. The figure reports category-frequency-level performance obtained from the semantic evidence fusion evaluation.

681 4.2. Semantic Fusion Strategy Comparison (RQ2)

682 This experiment investigates how various semantic evidence aggregation
 683 strategies influence the final fusion scores generated by the semantic evi-
 684 dence fusion framework. After generating semantic prompt evidence for each
 685 image-category pair, several fusion mechanisms were applied to combine the
 686 available evidence. Mean fusion calculates the average contribution of all
 687 semantic prompts, weighted fusion assigns varying importance values to the
 688 sources of semantic evidence, adaptive fusion dynamically adjusts evidence
 689 contributions based on confidence characteristics, and max fusion selects the
 690 strongest available semantic evidence. The goal was to determine which
 691 aggregation mechanism produces the most effective semantic evidence repre-
 692 sentation across the evaluated datasets and category frequency groups.

693 Table 5 presents an overall comparison of the performance of the evalu-
 694 ated fusion strategies. Max Fusion achieved the highest mean fusion score

of 0.3292, ranking first among all evaluated approaches. Adaptive Fusion attained a mean score of 0.2184, taking second place, while Weighted Fusion and Mean Fusion achieved mean scores of 0.1436 and 0.1420, respectively. A similar pattern is observed in the median values, with Max Fusion achieving the highest median at 0.3039, compared to 0.2061 for Adaptive Fusion. Furthermore, the standard deviations indicate that Max Fusion produced a greater spread of semantic evaluation scores, whereas the other strategies yielded lower mean fusion scores across the evaluation datasets. These rankings reflect the behavior of the fusion scores rather than official performance in the detection benchmark, as the comparison is based on aggregate statistics rather than AP50 proxies or full benchmark metrics.

Table 5: Performance comparison of the evaluated semantic evidence fusion strategies. Mean, median, and standard deviation of fusion scores are reported together with category-frequency-group statistics and overall ranking.

Strategy	Mean Score	Median Score	Std. Score	Rare Mean	Common Mean	Frequent Mean	Rank
Max Fusion	0.3292	0.3039	0.2184	0.1797	0.2405	0.3302	1
Adaptive Fusion	0.2184	0.2061	0.1381	0.1289	0.1637	0.2165	2
Weighted Fusion	0.1436	0.1275	0.1058	0.0949	0.1117	0.1395	3
Mean Fusion	0.1420	0.1259	0.1048	0.0947	0.1111	0.1377	4

Table 6 presents the analysis of category frequency groups for the evaluated fusion strategies. For rare categories, Max Fusion achieved a mean value of 0.1797, compared to 0.1289 for Adaptive Fusion and approximately 0.095 for both Weighted and Mean Fusion. For frequent categories, Max Fusion achieved a mean value of 0.2405, while Adaptive Fusion reached 0.1637. Frequent categories yielded the highest fusion values across all evaluated strategies, with Max Fusion achieving a mean of 0.3302 and Adaptive Fusion 0.2165. The ranking remained consistent across the rare, frequent, and very frequent groups, demonstrating similar fusion behavior throughout the long-tail category distribution.

Figure 8 shows the overall comparison of the rankings for the evaluated fusion strategies. The figure visualizes the differences between Max Fusion, Adaptive Fusion, Weighted Fusion, and Mean Fusion based on the average fusion scores achieved during the evaluation. There is a distinct performance gap between Max Fusion and the other approaches, whereas Weighted Fusion and Mean Fusion exhibit very similar score distributions. The figure provides a graphical representation of the general ranking behavior of the evaluated mechanisms for aggregated semantic evidence.

Table 6: Frequency-group analysis of strategies for semantic evidence fusion across the rare, frequent, and very frequent categories of LVIS. Fusion score statistics and the retrieval of semantic pairs are reported for each strategy tested.

Group	Strategy	GT Pairs	Mean Score	Median Score	Std. Score	Pair Recall	Rank
Rare	Max Fusion	331	0.1797	0.1493	0.1586	0.7674	1
Rare	Adaptive Fusion	331	0.1289	0.1189	0.1023	0.7674	2
Rare	Weighted Fusion	331	0.0949	0.0945	0.0728	0.7674	3
Rare	Mean Fusion	331	0.0947	0.0946	0.0725	0.7674	4
Common	Max Fusion	266	0.2405	0.2113	0.1735	0.8722	1
Common	Adaptive Fusion	266	0.1637	0.1566	0.1029	0.8722	2
Common	Weighted Fusion	266	0.1117	0.1094	0.0646	0.8722	3
Common	Mean Fusion	266	0.1111	0.1070	0.0644	0.8722	4
Frequent	Max Fusion	3535	0.3302	0.3090	0.2103	0.9298	1
Frequent	Adaptive Fusion	3535	0.2165	0.2072	0.1269	0.9298	2
Frequent	Weighted Fusion	3535	0.1395	0.1264	0.0898	0.9298	3
Frequent	Mean Fusion	3535	0.1377	0.1253	0.0888	0.9298	4

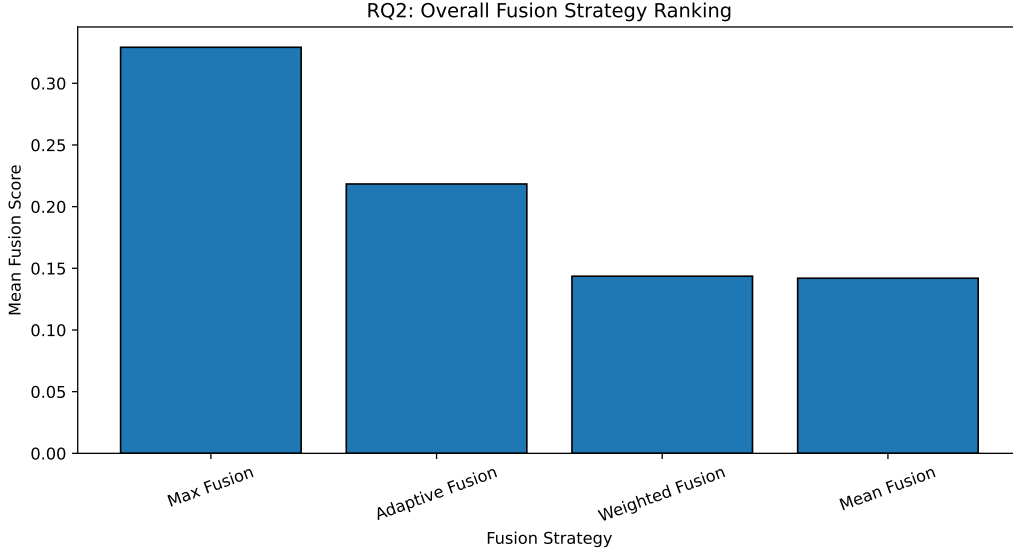


Figure 8: Comparison of the evaluated semantic evidence fusion strategies using the mean fusion scores obtained across the evaluation datasets. The figure reports the relative ranking of Mean Fusion, Weighted Fusion, Adaptive Fusion, and Max Fusion.

724 Figure 9 shows the threshold sensitivity analysis of the evaluated fusion
725 strategies. The figure illustrates how performance changes across the an-
726 analyzed range of the decision threshold and provides a comparison of the
727 strategies' stability under various threshold configurations. The plotted val-
728 ues show performance variations as the decision threshold changes, enabling

729 a comparison of the robustness of the evaluated fusion mechanisms under dif-
730 ferent operating conditions. The threshold sensitivity analysis was conducted
731 by varying the decision threshold across the evaluated operating range while
732 keeping all other fusion parameters constant. The threshold range was ex-
733 amined to determine a stable operating point for the aggregation of semantic
734 evidence. A threshold of 0.10 was subsequently selected for the remaining
735 experiments, as it offered a favorable balance between recall and false alarm
736 behavior across the evaluated datasets.

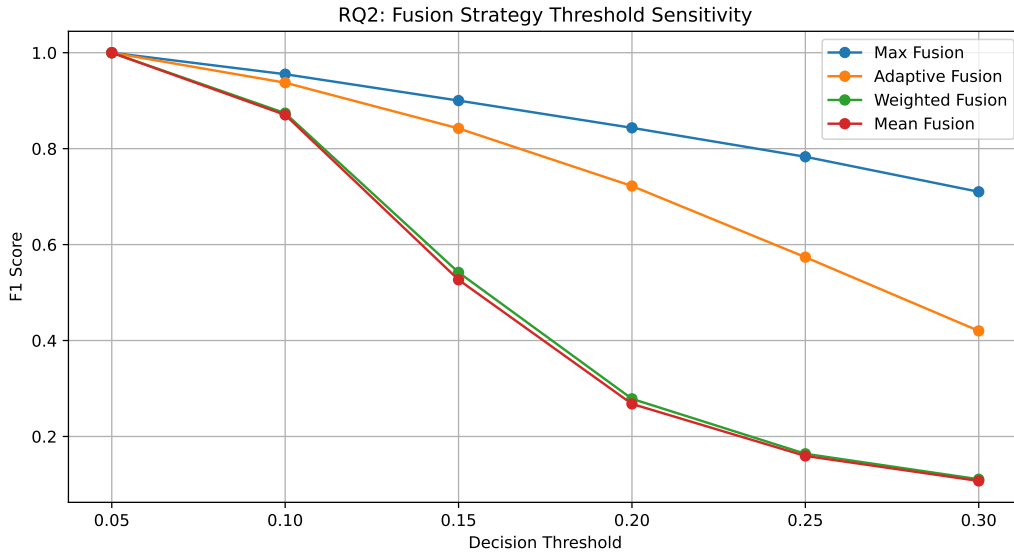


Figure 9: Threshold-sensitivity analysis of the evaluated semantic evidence fusion strategies. The figure reports the variation in performance across the analysed decision-threshold range.

737 4.3. Semantic Uncertainty Conflict Analysis (RQ3)

738 This experiment evaluates the uncertainty characteristics of semantic evi-
739 dence generated from multiple prompt sources. Following semantic evidence
740 fusion, additional uncertainty metrics were calculated using indicators for
741 entropy, semantic conflict, variation, reliability, uncertainty risk, and false-
742 positive risk. These metrics were computed for each image-category pair
743 to quantify the consistency of the semantic evidence derived from different

prompts. The aim was to determine whether uncertainty and semantic conflict metrics could be used to characterize the reliability of semantic evidence and to analyze its relationship with false-positive detections in the evaluated datasets.

Table 7 presents the dataset level uncertainty and false-positive risk statistics obtained through semantic evidence fusion. For the COCO subset, semantic evidence fusion achieved a recall of 0.9270 with 2,982 detected image-category pairs. The corresponding entropy, conflict, and variance values were 0.7179, 0.4487, and 0.5566, while the false-positive rate was 0.6765. For the LVIS subset, recall was 0.9131 with 3,773 detected pairs. The corresponding entropy, conflict, and variance values were 0.7423, 0.4434, and 0.5201, while the false-positive rate rose slightly to 0.6940. The results further demonstrate that, despite differing category distributions and prompt usage, both datasets yielded similar uncertainty risk values of approximately 0.58 and false-positive risk proxy values close to 1.8.

Table 7: Dataset-level uncertainty, conflict, reliability, and false-positive risk statistics obtained from semantic evidence fusion on the COCO and LVIS evaluation subsets.

Dataset	GT Pairs	Detected Pairs	Recall	Entropy	Conflict	Variation	Reliability	Uncertainty Risk	FP Risk Proxy	TP	FP	FP Rate	Prompt
COCO	3217	2982	0.9270	0.7179	0.4487	0.5566	0.1037	0.5834	1.7933	5108	29912	0.6765	10.8
LVIS	4132	3773	0.9131	0.7423	0.4434	0.5201	0.0872	0.5821	1.7567	6518	41352	0.6940	11.5

Table 8 presents the analysis of uncertainty-risk bins, created by grouping detections into low-, medium-, and high-risk intervals based on their uncertainty values. For the COCO subset, the false-positive rate rose from 0.5147 in the low-risk group to 0.7907 and 0.8819 in the medium and high-risk groups. Similarly, for the LVIS subset, the false-positive rate increased from 0.5832 in the low-risk group to 0.8161 and 0.8821 in the medium- and high-risk groups. Corresponding measurements for entropy, conflict, variation, and false-positive risk also increased across the uncertainty groups. Additionally, the average number of prompts rose from approximately 3 prompts in the low-risk groups to about 19 prompts in the high-risk groups, indicating greater semantic complexity in the higher-risk detections.

Figure 10 shows the behavior of false alarms across the evaluated uncertainty risk intervals. The figure visualizes how false alarm rates progress from low risk to high risk for both COCO and LVIS. A consistent increase in the false alarm rate is observed as the uncertainty risk rises, with the highest false alarm rates occurring in the high-risk intervals for both datasets. The figure provides a graphical representation of the uncertainty bin analysis and

Table 8: Uncertainty-risk-bin analysis obtained from semantic evidence fusion. Entropy, conflict, reliability, false-positive risk, and detection statistics are reported for low-, medium-, and high-risk uncertainty groups.

Dataset	Risk Bin	Pairs	Entropy	Conflict	Variation	Reliability	Uncertainty Risk	FP Risk Proxy	TP	FP	FP Rate	Prompt Count
COCO	Low Risk	993	0.5768	0.1887	0.2438	0.1684	0.3577	0.3704	1094	2186	0.5147	3.3031
COCO	Medium Risk	976	0.8861	0.5439	0.6273	0.0898	0.7016	1.8758	2128	10438	0.7907	12.8750
COCO	High Risk	1013	0.8607	0.7159	0.9241	0.0778	0.8259	3.5246	1886	17288	0.8819	18.9279
LVIS	Low Risk	1259	0.6576	0.2033	0.2440	0.1223	0.3952	0.4030	1369	3199	0.5832	3.6283
LVIS	Medium Risk	1275	0.9037	0.5437	0.5925	0.0872	0.6999	1.9183	2706	15875	0.8161	14.5733
LVIS	High Risk	1239	0.8774	0.7126	0.8768	0.0768	0.8195	3.4751	2443	22278	0.8821	19.9524

776 highlights the relationship between uncertainty measurements and detection
777 quality.

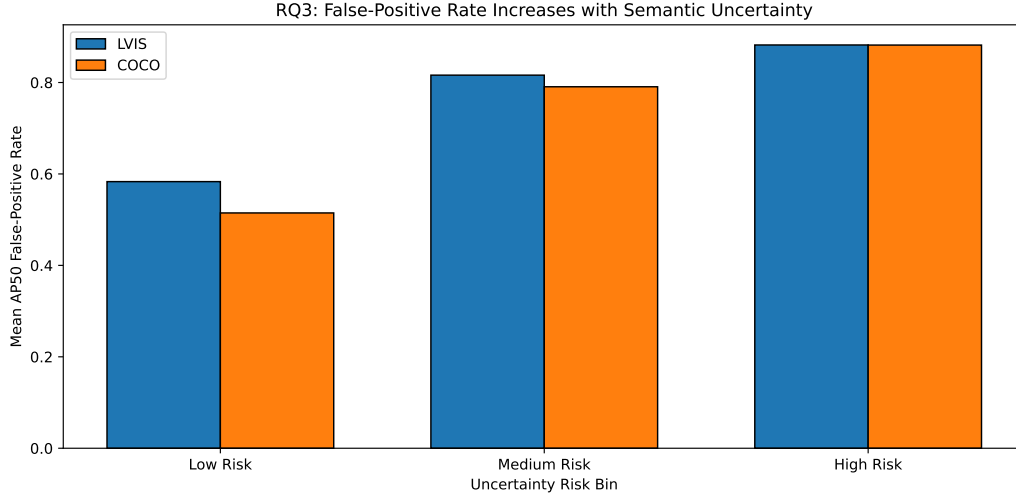


Figure 10: False-positive behaviour across the evaluated uncertainty-risk intervals for the COCO and LVIS evaluation subsets. The figure reports the false-positive rates observed within the low-, medium-, and high-risk uncertainty groups.

778 Figure 11 illustrates the relationship between semantic conflict measure-
779 ments and the number of false alarms obtained through semantic evidence
780 fusion. The figure visualizes conflict levels and the corresponding false alarm
781 statistics for the evaluated datasets. The dataset-level comparison suggests
782 a potential link between semantic conflict and false alarm behavior; however,
783 further experiments would be required to confirm a stronger causal relation-
784 ship within the evaluated semantic evidence fusion framework. The figure

785 provides a dataset-level view of how semantic conflict influences false alarm
786 behavior within the semantic evidence fusion framework. The observed trend
787 suggests that increasing disagreement among semantic prompt sources may
788 indicate unreliable semantic evidence, thereby motivating the use of conflict-
789 aware diagnostic measures within the fusion framework.

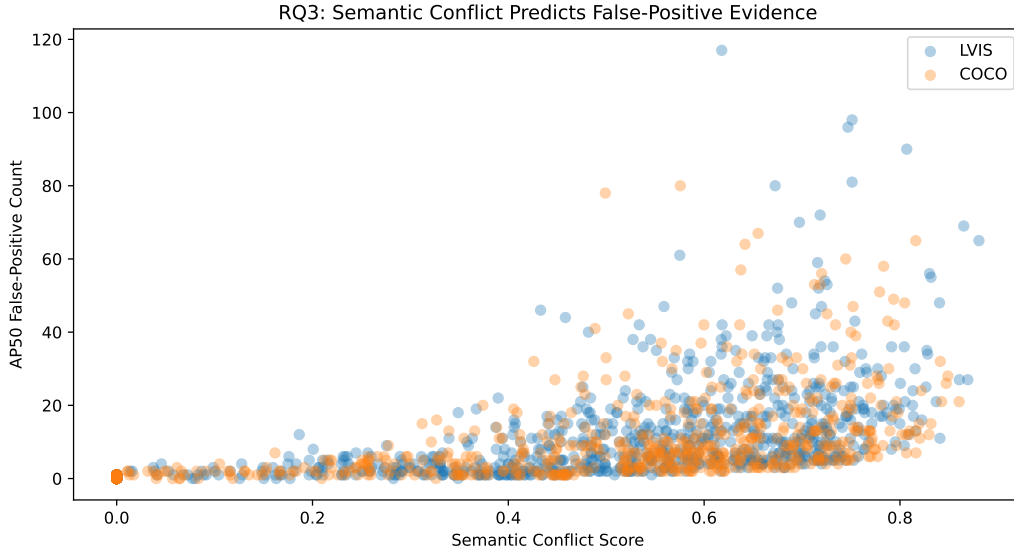


Figure 11: Relationship between semantic conflict measurements and false-positive counts obtained from semantic evidence fusion. The figure reports dataset-level conflict and false-positive statistics for the evaluated datasets.

790 4.4. Semantic Redundancy Complementarity Analysis (RQ4)

791 This experiment investigates how various semantic prompt categories contribute to the overall performance of the semantic evidence fusion framework.
792 A prompt ablation analysis was conducted by systematically removing individual prompt categories and re-evaluating semantic pair recall. The aim
793 was to measure the contribution of each prompt source to semantic evidence generation and to determine whether prompt categories provide unique semantic
794 information or contain redundant evidence already available from other prompt sources. Additionally, redundancy and complementarity statistics
795 were calculated to quantify the amount of shared and unique semantic
796 information. The results of this analysis are presented in the following sections.
797
798
799

evidence provided by the evaluated prompt categories across datasets and category frequency groups.

Table 9 presents the results of the prompt ablation performed after removing individual semantic prompt categories from the fusion framework. Descriptive prompts achieved the highest coverage rate of 0.5682; their removal resulted in the largest drop in recall 0.2551 representing a relative decline of 27.76%. Primary prompts led to a recall decrease of 0.0404, while context and attribute prompts resulted in reductions of 0.0158 and 0.0150, respectively. Hypernym and synonym prompts caused the smallest declines in recall, at 0.0083 and 0.0082, respectively. The unique contribution rate was also highest for descriptive prompts (0.2551), indicating that a significant portion of the semantic cues originated from descriptive prompt formulations that could not be recovered by the remaining prompt categories.

Table 9: Prompt-ablation analysis of semantic prompt categories. Coverage, contribution, and recall-reduction statistics are reported for the evaluated prompt sources.

Prompt Type	Coverage	Unique Rate	Redundant Rate	Recall (All)	Recall (Removed)	Recall Drop	Drop (%)	Rank
Description	0.5682	0.2551	0.3131	0.9192	0.6640	0.2551	27.76	1
Primary	0.4074	0.0404	0.3670	0.9192	0.8788	0.0404	4.40	2
Context	0.3937	0.0158	0.3779	0.9192	0.9034	0.0158	1.72	3
Attribute	0.3237	0.0150	0.3087	0.9192	0.9042	0.0150	1.63	4
Hypernym	0.1159	0.0083	0.1076	0.9192	0.9109	0.0083	0.90	5
Synonym	0.1826	0.0082	0.1744	0.9192	0.9110	0.0082	0.89	6

Table 10 presents the analysis of redundancy and complementarity across datasets and LVIS frequency groups. For the COCO subset, 1,300 image-category pairs were supported by only a single prompt source, while 1,682 pairs were supported by multiple prompt sources, resulting in a redundancy rate of 0.5228. For the LVIS subset, the number of redundant pairs rose to 2,554, corresponding to a redundancy rate of 0.6181. Within the LVIS frequency groups, redundancy rates increased from 0.4743 for rare categories to 0.5714 for common categories and up to 0.6351 for frequent categories. The average number of prompt types contributing to each detected pair also increased from 1.5650 for rare categories to 2.2065 for frequent categories, indicating greater overlap in semantic evidence for categories with higher representation.

Figure 12 shows the reduction in recall that occurs when individual semantic prompt categories are removed from the fusion framework. The figure illustrates the relative importance of each prompt category and highlights the significant contribution of description prompts compared to the other prompt sources. A clear gap is evident between description prompts and all other

Table 10: Redundancy and complementarity summary across datasets and LVIS frequency groups. The table reports semantic recall, single-source contribution, redundancy, missed pairs, and mean prompt-type utilization.

Level	Group	GT Pairs	Detected Pairs	Recall	Single-Source Pairs	Redundant Pairs	Missed Pairs	Single-Source Rate	Redundancy
Dataset	COCO	3217	2982	0.9270	1300	1682	235	0.4041	0.5218
Dataset	LVIS	4132	3773	0.9131	1219	2554	359	0.2950	0.6181
LVIS Frequency	Rare	331	254	0.7674	97	157	77	0.2931	0.4771
LVIS Frequency	Common	266	232	0.8722	80	152	34	0.3008	0.5718
LVIS Frequency	Frequent	3535	3287	0.9298	1042	2245	248	0.2948	0.6331

categories, whereas hypernym and synonym prompts result in the smallest reductions in recall for semantic pairs. The figure provides a graphical representation of the prompt ablation analysis and the relative contribution of each semantic source of evidence.

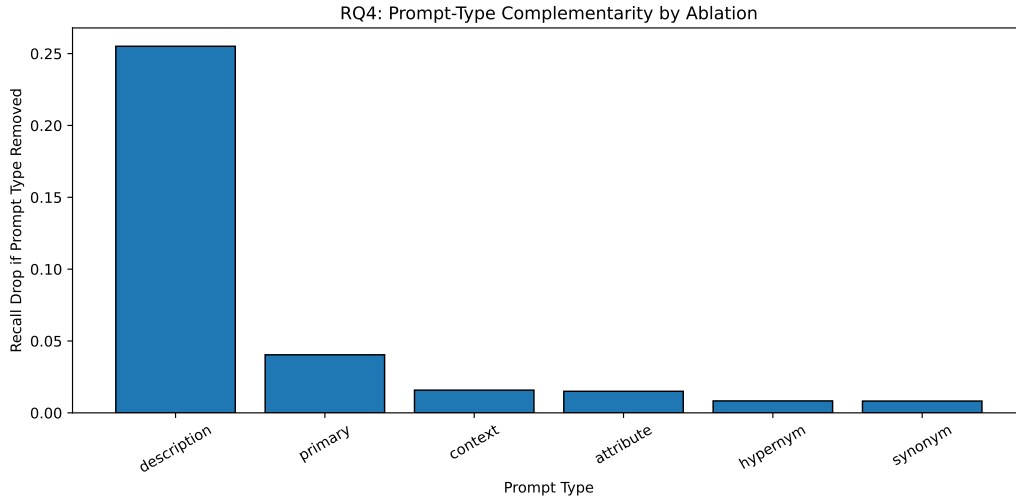


Figure 12: Recall reduction observed after removing individual semantic prompt categories from the semantic evidence fusion framework. The figure reports the relative contribution of each prompt source to overall semantic pair recall.

The results show that semantic complementarity contributes significantly more to overall semantic recall than semantic redundancy. Although redundant prompt sources ensure robustness through repeated cues, prompt ablation analysis reveals that unique semantic information particularly from descriptive prompts—is responsible for the greatest improvements in recall within the fusion framework.

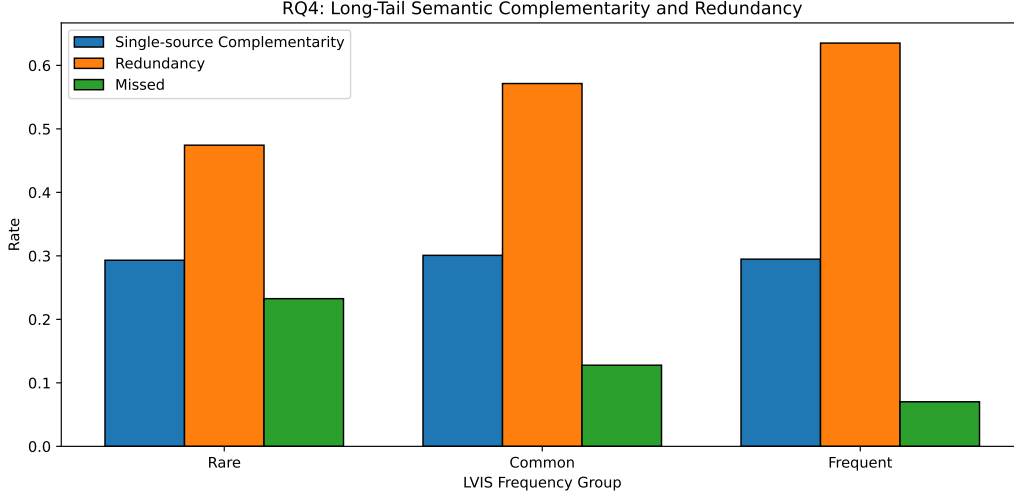


Figure 13: Redundancy and complementarity measurements obtained across the evaluated datasets and LVIS frequency groups. The figure reports the distribution of single-source and redundant semantic evidence contributions.

840 4.5. Prior-Aware Semantic Fusion Evaluation (RQ5)

841 This experiment investigates whether category frequency information can
842 be incorporated into semantic evidence fusion to improve object detection
843 performance within a long-tail distribution. Frequency-aware prior weights
844 were assigned to the LVIS categories rare, frequent, and very frequent and
845 integrated into the semantic evidence fusion process. The resulting "Prior-
846 Aware Fusion" results were compared with "Adaptive Fusion" to determine
847 whether knowledge of category frequencies could enhance the utilization of se-
848 mantic evidence and boost detection performance for underrepresented cate-
849 gories. Furthermore, the fusion methods were compared to benchmark Prior-
850 Aware Fusion against alternative aggregation mechanisms and to evaluate the
851 impact of frequency-aware prior weighting on the performance of semantic
852 evidence fusion.

853 Table 11 shows the performance comparison between Adaptive Fusion
854 and Prior-Aware Fusion across the evaluated frequency groups. For rare cat-
855 egories, recall increased from 0.7638 to 0.9409 after incorporating category
856 frequency priors, representing an absolute gain of 0.1772 and a relative im-
857 provement of 23.20%. For frequent categories, recall values were 0.8578 for
858 Adaptive Fusion and 0.9310 for Prior-Aware Fusion, corresponding to an ab-

859 solute gain of 0.0733 and a relative improvement of 8.54%. Very frequent
860 categories achieved recall values of 0.8880 and 0.9191, representing an abso-
861 lute gain of 0.0310 and a relative improvement of 3.49%. Thus, the greatest
862 improvement was observed for rare categories, while the gains for frequent
863 and very frequent categories were progressively smaller. This behaviour is
864 consistent with the purpose of frequency-aware weighting, since rare cate-
865 gories receive stronger prior support during semantic evidence aggregation
866 than common or frequent categories.

Table 11: Performance comparison between Adaptive Fusion and Prior-Aware Fusion across the LVIS frequency groups. Recall improvements obtained after incorporating category-frequency priors are reported.

Frequency Group	GT Pairs	Adaptive Recall	Prior-Aware Recall	Absolute Gain	Relative Gain (%)
Rare	331	0.7638	0.9409	0.1772	23.20
Common	266	0.8578	0.9310	0.0733	8.54
Frequent	3535	0.8880	0.9191	0.0310	3.49

867 Table 12 presents a comparison of the fusion methods used to evaluate the
868 impact of prior weightings on the performance of semantic evidence fusion.
869 Prior-Aware Fusion achieved the highest overall recall of 0.9213, ranking first
870 among the evaluated approaches. Max Fusion achieved an overall recall of
871 0.9109, taking second place, while Adaptive Fusion achieved an overall recall
872 of 0.8778, ranking third. Max Fusion yielded the highest average fusion score
873 of 0.3124 and the highest median value of 0.2870, whereas Prior-Aware Fusion
874 achieved average and median values of 0.2396 and 0.2287, respectively. The
875 results indicate that incorporating category frequency priors improved overall
876 recall while maintaining competitive fusion score performance.

877 It should be noted that the comparison reported in Table 12 evaluates
878 only the fusion based decision strategies under a uniform threshold protocol.
879 Consequently, the results are not directly comparable to the level of semantic
880 evidence reported in RQ6, which represents the unconstrained accumulation
881 of semantic evidence prior to adaptive or prior-driven decision filtering.

Table 12: Performance comparison of the evaluated fusion methods. Overall recall and fusion-score statistics are reported for Prior-Aware Fusion, Max Fusion, and Adaptive Fusion.

Method	Overall Recall	Mean Score	Median Score	Std. Score	Rank
Prior-Aware Fusion	0.9213	0.2396	0.2287	0.1450	1
Max Fusion	0.9109	0.3124	0.2870	0.2093	2
Adaptive Fusion	0.8778	0.2061	0.1975	0.1264	3

Figure 14 compares the recall values of Adaptive Fusion and Prior-Aware Fusion across LVIS frequency groups. The figure illustrates the recall improvements achieved by incorporating category frequency priors and highlights the differences between rare, frequent, and very frequent categories. The largest difference between the two fusion approaches occurs in the rare category group, whereas only minor improvements are observed for frequent and very frequent categories. The figure provides a graphical representation of the performance gains in the long tail achieved through frequency-based semantic evidence fusion.

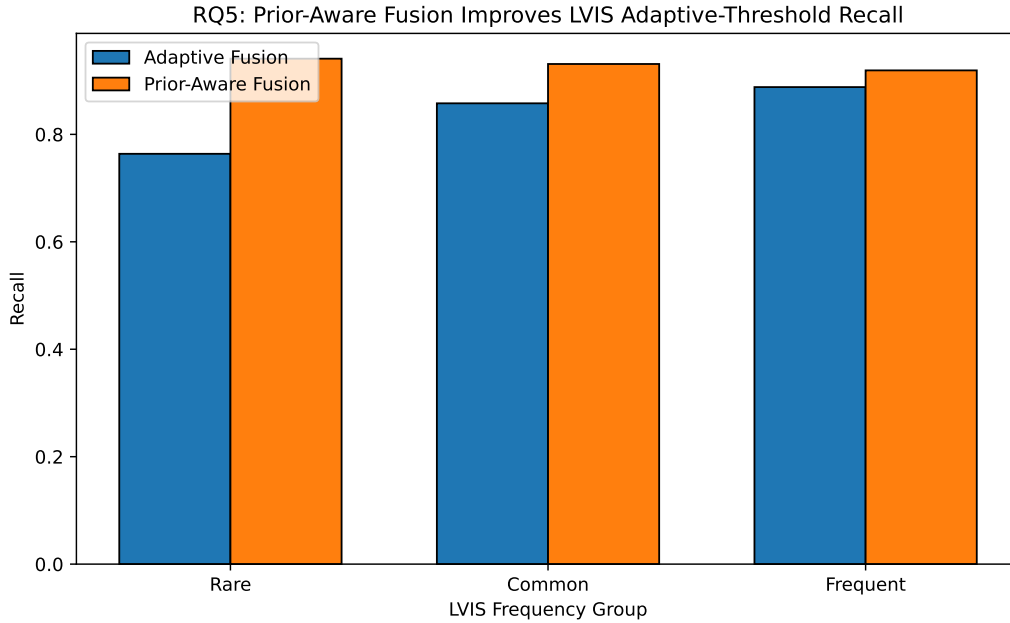


Figure 14: Recall comparison between Adaptive Fusion and Prior-Aware Fusion across the LVIS rare, common, and frequent category groups. The figure reports the recall improvements obtained after incorporating category-frequency priors.

Figure 15 shows the comparative performance of Prior-Aware Fusion, Max Fusion, and Adaptive Fusion based on overall recall and fusion score statistics. The figure illustrates the behavior of overall recall and fusion score for Prior-Aware Fusion, Max Fusion, and Adaptive Fusion. Prior-Aware Fusion achieved the highest overall recall among the evaluated methods, while Max Fusion yielded the highest average and median fusion scores. The figure

897 ure provides a graphical summary of the trade-off between the magnitude of
898 the fusion score and overall detection performance for the fusion approaches
899 investigated.

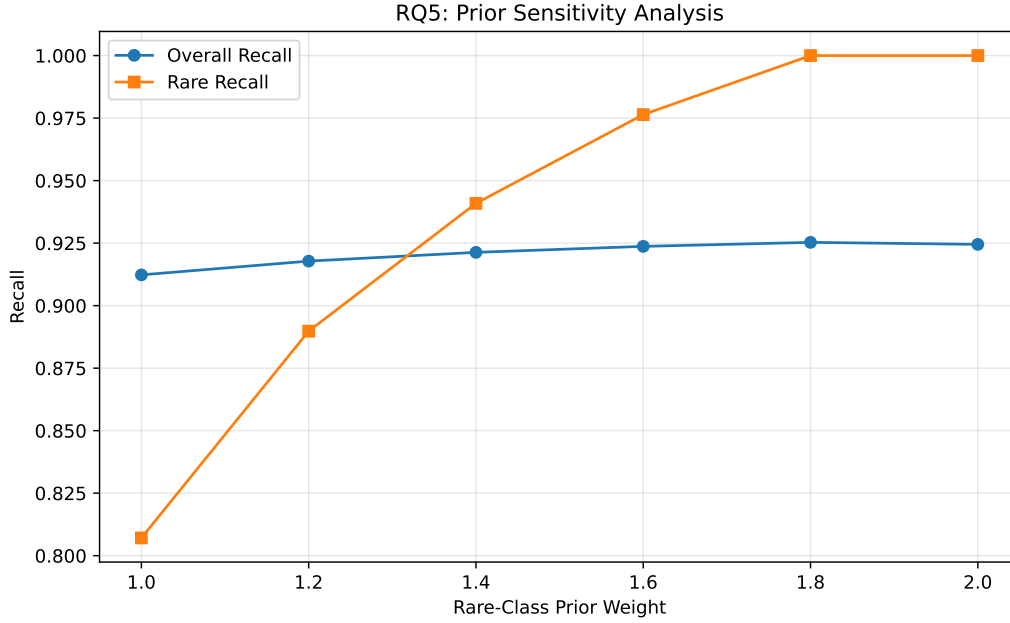


Figure 15: Performance comparison of the evaluated fusion methods based on overall recall and fusion-score statistics. The figure summarizes the results of the fusion-method comparison.

900 4.6. Framework Evolution Analysis (RQ6)

901 This experiment evaluates the entire semantic evidence fusion framework
902 by analyzing the progression from primary prompt detection through se-
903 mantic evidence fusion and adaptive fusion to prior-aware fusion. The goal
904 was to examine how the processing of semantic evidence changes across the
905 framework and to determine each stage’s contribution to overall detection
906 performance. All framework stages were evaluated using a common experi-
907 mental protocol on COCO and LVIS subsets, and additional analyses were
908 conducted across the LVIS categories of "rare," "frequent," and "very fre-
909 quent" to assess long-tail behavior.

910 RQ6 compares the stages across datasets and groups but does not involve
 911 repeated random restarts or statistical robustness tests. Your RQ6 table
 912 shows that semantic evidence fusion achieves the highest hit rate, whereas the
 913 adaptive and prior-aware stages are more selective. Maintain the explanation
 914 that this reflects the behavior and consistency of the framework stages rather
 915 than formal stochastic stability.

916 This experiment provides a consolidated overview of how successive
 917 evidence-processing mechanisms alter semantic evidence coverage and selec-
 918 tivity across datasets and groups with varying category frequencies. Unlike
 919 RQ1, which evaluates semantic evidence coverage, the Adaptive Fusion and
 920 Prior-Aware Fusion stages employ stricter mechanisms for evidence selection
 921 and confidence control. Consequently, these stages can reduce the retrieval
 922 rate while enhancing evidence selectivity and robustness.

923 The framework’s stability is assessed operationally as the consistency of
 924 retrieval behavior across datasets and groups with varying category frequen-
 925 cies, rather than through repeated stochastic trials. The observed ranking
 926 of the framework stages remained consistent across both COCO and LVIS
 927 subsets, indicating stable semantic evidence behavior under varying category
 928 structures.

929 Table 13 shows the comparison of the different framework stages at the
 930 dataset level. For the LVIS subset, recall increased from 0.8538 for Primary
 931 Prompt Detection to 0.9131 for Semantic Evidence Fusion, while the num-
 932 ber of detected image-category pairs rose from 3,528 to 3,773. Adaptive
 933 Fusion and Prior-Aware Fusion achieved recall values of 0.8015 and 0.8412,
 934 respectively. For the COCO subset, recall increased from 0.8816 for Primary
 935 Prompt Detection to 0.9270 for Semantic Evidence Fusion, while the number
 936 of detected pairs rose from 2,836 to 2,982. Adaptive Fusion and Prior-Aware
 937 Fusion both achieved a recall of 0.8231 with 2,648 detected pairs. Across
 938 both datasets, Semantic Evidence Fusion achieved the highest values for re-
 939 call and the average semantic evidence score among the evaluated framework
 940 stages. Consequently, the comparison of the framework stages reveals shifts
 941 in semantic coverage and evidence selectivity rather than a strictly monotonic
 942 improvement across all stages.

943 Table 14 presents the category-frequency group analysis of the framework
 944 development process. For rare categories, recall increased from 0.6616 for
 945 Primary Prompt Detection to 0.7674 for Semantic Evidence Fusion, while
 946 Prior Knowledge Fusion achieved a recall of 0.7221, compared to 0.5861 for
 947 Adaptive Fusion. For frequent categories, recall values were 0.8233, 0.8722,

Table 13: Dataset-level comparison of the evaluated framework stages. Recall, mean semantic evidence score, and detected image-category pairs are reported for the LVIS and COCO evaluation subsets.

Dataset	Framework Stage	GT Pairs	Recall	Mean Score	Detected Pairs
LVIS	Primary Prompt	4132	0.8538	0.2312	3528
LVIS	Semantic Evidence	4132	0.9131	0.3124	3773
LVIS	Adaptive Fusion	4132	0.8015	0.2061	3312
LVIS	Prior-Aware Fusion	4132	0.8412	0.2396	3476
COCO	Primary Prompt	3217	0.8816	0.2360	2836
COCO	Semantic Evidence	3217	0.9270	0.3508	2982
COCO	Adaptive Fusion	3217	0.8231	0.2341	2648
COCO	Prior-Aware Fusion	3217	0.8231	0.2341	2648

0.7481, and 0.8120 for Primary Prompt Detection, Semantic Evidence Fusion, Adaptive Fusion, and Prior Knowledge Fusion, respectively. Very frequent categories achieved the highest overall recall values, with Semantic Evidence Fusion attaining a recall of 0.9298. The results demonstrate that the relative performance ranking observed at the dataset level remains consistent across the LVIS frequency groups.

Table 14: Category-frequency-group comparison of the evaluated framework stages. Recall measurements are reported for the COCO dataset and the LVIS rare, common, and frequent category groups.

Dataset	Group	GT Pairs	Primary Recall	Semantic Recall	Adaptive Recall	Prior-Aware Recall
COCO	Balanced	3217	0.8816	0.9270	0.8231	0.8231
LVIS	Rare	331	0.6616	0.7674	0.5861	0.7221
LVIS	Common	266	0.8233	0.8722	0.7481	0.8120
LVIS	Frequent	3535	0.8741	0.9298	0.8257	0.8546

Figure 16 illustrates the dataset-specific evolution of the evaluated framework stages. The figure visualizes changes in retrieval and semantic evidence metrics as the framework progresses from Primary Prompt Detection through Semantic Evidence Fusion and Adaptive Fusion to Prior-Aware Fusion. The most significant performance boost is observed during the transition from Primary Prompt Detection to Semantic Evidence Fusion, while subsequent fusion stages modify the behavior of the semantic evidence through additional evidence aggregation and prior-weighting mechanisms. The figure provides a graphical overview of the framework’s entire development process for the COCO and LVIS evaluation subsets.

Figure 17 shows a comparison of category frequency groups across the evaluated framework stages. The figure illustrates the framework’s behavior regarding the rare, frequent, and common LVIS category groups, along-

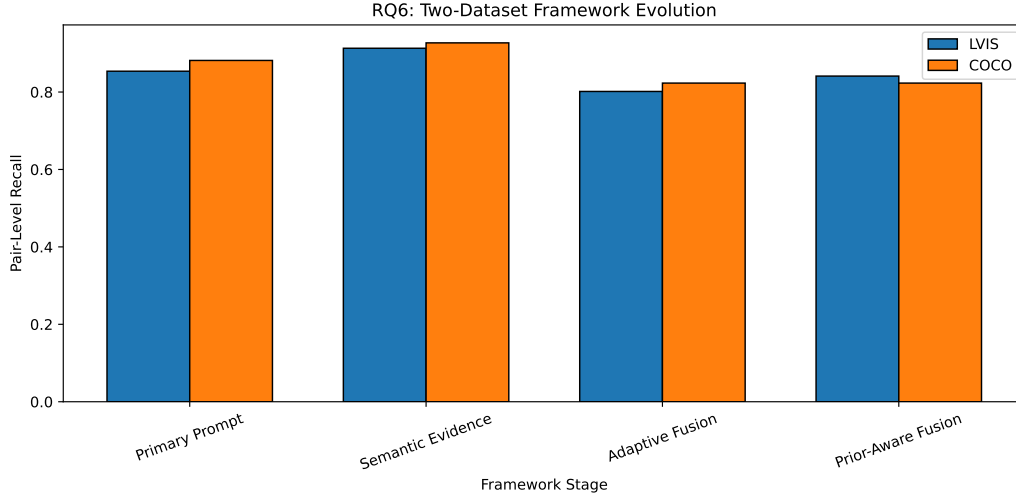


Figure 16: Dataset-level comparison of the evaluated framework stages on the LVIS and COCO evaluation subsets. The figure reports recall and semantic evidence measurements obtained across the framework evolution process.

side the COCO evaluation subset. Semantic Evidence Fusion consistently achieves the highest recall values across all evaluated groups, while Prior-Aware Fusion offers improvements over Adaptive Fusion, particularly for categories with lower representation. The figure summarizes the framework’s stability across different category frequency distributions and demonstrates performance progression throughout the Semantic Evidence Fusion pipeline.

Overall, the analysis of the framework’s development shows that Semantic Evidence Fusion offers the highest semantic coverage and best recall across both datasets, whereas Adaptive Fusion introduces a stricter evidence selection process that reduces recall in favor of more conservative decision-making. Prior-Aware Fusion partially compensates for this loss by incorporating category frequency information, yielding the greatest benefits for long-tail categories. These results demonstrate that semantic evidence generation is the primary driver of improved recall, while adaptive and prior-driven fusion mechanisms primarily influence evidence selection and stability.

4.7. Summary of Findings

The results addressing the six research questions provide a consolidated evaluation of the proposed semantic evidence fusion framework. RQ1 demon-

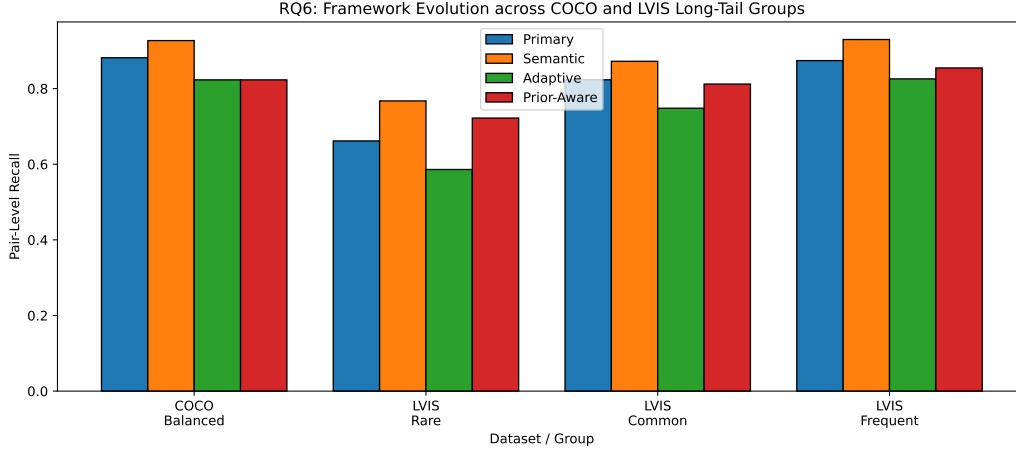


Figure 17: Category-frequency-group comparison of the evaluated framework stages. The figure reports recall measurements for the LVIS rare, common, and frequent category groups together with the COCO evaluation subset.

985 strated that semantic evidence fusion improved recall and AP50 proxy per-
 986 formance compared to primary prompt detection on both the COCO and
 987 LVIS evaluation subsets. RQ2 showed that Max Fusion achieved the high-
 988 est fusion score statistics, whereas Adaptive, Weighted, and Mean Fusion
 989 produced lower score distributions. RQ3 revealed that uncertainty-risk re-
 990 gions were associated with increasing false-positive rates across both datasets.
 991 RQ4 indicated that description prompts provided the greatest unique se-
 992 mantic evidence contribution among the evaluated prompt categories. RQ5
 993 showed that Prior-Aware Fusion achieved the largest recall improvement for
 994 rare LVIS categories. RQ6 demonstrated that the behavior of the framework
 995 stages varied depending on the dataset and category frequency group; se-
 996 mantic evidence fusion achieved the highest recall, while later fusion stages
 997 altered the evidence selectivity.

998 5. Discussion

999 The aim of this study was to investigate whether multiple semantically
 1000 related prompts could be fused as correlated sources of semantic evidence to
 1001 support the detection of rare objects within a vision-language model frame-
 1002 work. The experimental results presented in the previous section demon-

1003 strate that the fusion of semantic evidence influences detection performance
1004 in both the LVIS and COCO evaluation settings, resulting in measurable
1005 changes in recall, localization performance, uncertainty characteristics, and
1006 behavior related to category frequency.

1007 The results regarding RQ1 show that fusing semantic evidence improves
1008 recall for image categories compared to detection using a primary prompt
1009 across both datasets. The observed improvements suggest that semantically
1010 related prompts provide additional semantic evidence that extends beyond
1011 the information contained in a single category description. The increase
1012 in semantic detections and recall values indicates that diversifying semantic
1013 prompts expands semantic coverage, enabling the identification of additional
1014 object-category relationships during the detection process. The most sig-
1015 nificant improvements were observed on the LVIS dataset, which exhibits a
1016 more pronounced long-tail category distribution. Although Semantic Evi-
1017 dence Fusion achieved the highest recall, more selective evidence aggrega-
1018 tion mechanisms were intentionally employed in later fusion stages. These
1019 stages trade off recall for improved evidence filtering, uncertainty control,
1020 and frequency-aware behavior.

1021 The RQ2 experiments investigated alternative fusion strategies for aggre-
1022 gating semantic evidence. The evaluated fusion mechanisms produced vary-
1023 ing characteristics in the semantic evidence and score distributions, demon-
1024 strating that evidence aggregation influences the behavior of the detection
1025 framework. The results show that semantic evidence from multiple sources
1026 can be combined via various aggregation mechanisms while maintaining con-
1027 sistent scoring across datasets. The comparison also highlights the impor-
1028 tance of selecting an appropriate aggregation strategy when multiple sources
1029 of semantic evidence are available.

1030 The analysis of uncertainty and semantic conflicts conducted in RQ3 pro-
1031 vides further insight into the behavior of semantic evidence. The occurrence
1032 of uncertainty and semantic conflicts within the evaluated datasets demon-
1033 strates that different prompt sources do not always yield identical evidence.
1034 Variability among sources of semantic evidence is therefore a fundamental
1035 characteristic of prompt-based object detection. Furthermore, uncertainty
1036 analysis demonstrates that the reliability of semantic evidence can be as-
1037 sessed using information derived directly from the fusion process, rather than
1038 relying solely on the final detection results.

1039 The analysis of redundancy and complementarity conducted in RQ4 re-
1040 veals that individual semantic prompt categories contribute differently to the

fusion of semantic evidence. Prompt ablation experiments have shown measurable changes in recall following the removal of specific sources of semantic evidence, indicating that certain prompt categories provide unique information not fully captured by other prompts. These observations support the use of multiple semantic prompt formulations within the proposed framework and demonstrate that the diversity of semantic evidence contributes to overall detection coverage.

The results from RQ5 indicate that category frequency information influences semantic evidence fusion behavior in the long-tail setting. Incorporating frequency-aware prior information led to measurable changes across LVIS frequency groups, demonstrating that category frequency characteristics can be directly integrated into the semantic evidence aggregation process. This observation is particularly relevant for long-tail object detection, as category frequency imbalance remains one of the major challenges in detecting rare categories.

Finally, the framework-level analysis conducted in RQ6 provides a comprehensive overview of the semantic evidence fusion pipeline. The comparison between primary prompt detection, semantic evidence fusion, adaptive fusion, and prior-aware fusion illustrates how semantic evidence evolves throughout the framework. The results indicate that semantic prompt generation, semantic evidence aggregation, uncertainty analysis, and the integration of category frequencies can be combined within a unified workflow for long-tail object detection. Furthermore, the framework’s consistency across both datasets demonstrates the applicability of semantic evidence fusion to varying object category distributions.

Overall, the results demonstrate that semantic prompt generation and semantic evidence aggregation can be formulated as an information fusion problem, in which multiple semantically related prompts serve as complementary sources of evidence. The proposed framework offers a structured method for analyzing the behavior of semantic evidence and provides empirical insights into the role of semantic diversity in long-tail object recognition based on vision and language.

5.1. Limitations

Although the proposed framework demonstrated consistent behavior regarding semantic evidence fusion across the evaluated datasets, certain limitations should be acknowledged. First, the experiments were conducted using curated subsets of the LVIS and COCO datasets rather than full benchmark

evaluations. Second, the study focused on semantic evidence behavior using the OWL-ViT vision-language model; reported results might differ if alternative vision-language architectures were employed. Third, the framework examined semantic prompt generation and evidence aggregation without modifying detector parameters, thereby limiting the analysis to semantic evidence fusion at inference time. Finally, fusion was evaluated with respect to category frequency using LVIS frequency groups and might exhibit different behavior if applied to datasets with alternative category distributions.

5.2. Future Directions

Future research could investigate how to fuse semantic evidence with additional vision-language architectures to evaluate the generalizability of the proposed framework. Automated semantic prompt generation using large language models could also increase semantic diversity and coverage for rare object categories. Advanced strategies for aggregating semantic evidence—including uncertainty-aware fusion and adaptive prompt selection mechanisms could likewise be explored to improve evidence integration. Furthermore, the proposed methodology could be extended beyond object detection to other multi-modal information fusion applications, such as visual grounding, semantic scene understanding, image retrieval, and vision-language reasoning systems.

6. Conclusion

This study investigated the fusion of semantic evidence for long-tail object detection using vision-language models and information fusion principles. The proposed framework treated multiple semantically related prompts as correlated sources of semantic evidence and integrated them via a semantic evidence fusion pipeline implemented using OWL-ViT. The framework encompassed semantic prompt generation, semantic evidence aggregation, uncertainty and conflict analysis, the assessment of redundancy and complementarity, and frequency-aware prior modeling within a unified object detection workflow.

The first research question examined whether multiple semantic prompts could improve object detection performance compared to detection based on a primary prompt. The results showed that semantic evidence fusion improved image category recall and AP50 proxy performance on both the LVIS and COCO evaluation sets. The observed improvements suggest that

1113 semantically related prompts provide additional semantic coverage beyond
1114 the description of a single category.

1115 The second research question investigated alternative strategies for aggre-
1116 gating semantic evidence. Several fusion mechanisms were tested, including
1117 mean fusion, max fusion, weighted fusion, adaptive fusion, and prior-aware
1118 fusion. The results demonstrated that semantic evidence can be aggregated
1119 in various ways and that the aggregation behavior influences the resulting
1120 detection performance.

1121 The third research question investigated characteristics of uncertainty
1122 and semantic conflict within the fusion process. The analysis demonstrated
1123 that measures of uncertainty and conflict can be derived directly from the
1124 behavior of semantic evidence and provide additional information regarding
1125 detection reliability and behavior in the event of false alarms.

1126 The fourth research question examined redundancy and complementar-
1127 ity among the sources of semantic cues. Cue-omission experiments revealed
1128 that different cue categories contribute differently to the fusion of seman-
1129 tic evidence and that complementary semantic information exists across the
1130 evaluated cue sources.

1131 The fifth research question evaluated the integration of category frequency
1132 information via prior-aware fusion. The results demonstrated that frequency-
1133 aware semantic aggregation can be incorporated into the semantic evidence
1134 fusion process and influences performance across long-tail category groups.

1135 The final research question examined the further development and sta-
1136 bility of the framework. The comparison among primary prompt detection,
1137 semantic evidence fusion, adaptive fusion, and prior-aware fusion demon-
1138 strated the progression of semantic evidence throughout the framework and
1139 provided a unified evaluation of the proposed approach across datasets and
1140 category frequency groups.

1141 The main contribution of this study is the formulation of semantic prompt
1142 aggregation as an information fusion problem for long-tail object detection.
1143 The proposed framework offers a reproducible methodology for generating,
1144 aggregating, and analyzing semantic cues within vision-language object de-
1145 tection systems. Furthermore, the study provides empirical evidence regard-
1146 ing the behavior of semantic cues, uncertainty characteristics, prompt com-
1147 plementarity, and category frequency effects in long-tail object detection.

1148 Certain limitations should be acknowledged. Evaluation was conducted
1149 on curated subsets of the LVIS and COCO datasets rather than through
1150 full benchmark evaluations. Moreover, the study focused on the behavior

of semantic cues within a prompt-based OWL-ViT framework and did not investigate detector retraining or optimization on large-scale benchmarks. These limitations present opportunities for future research.

Future work could investigate semantic evidence fusion using additional vision-language architectures, larger evaluation datasets, automated techniques for semantic prompt generation, and advanced evidence aggregation mechanisms. Further research could also explore uncertainty-aware semantic fusion, adaptive prompt selection, and multi-modal evidence integration for rare object detection.

In summary, the results demonstrate that semantic evidence fusion offers a practical information fusion framework for integrating multiple semantically related prompts within vision-language object detection systems. The proposed methodology provides new insights into the behavior of semantic evidence and establishes a foundation for future research on semantic information fusion for rare object detection.

Declarations

The authors used the generative AI tool ChatGPT (OpenAI, San Francisco, CA, USA) to improve the language and clarity of the manuscript. The authors reviewed and edited all content generated by the tool and take full responsibility for the final version of the manuscript.

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